

Manufacturing and Service Operations

MSO 40.012

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Summary Course 4

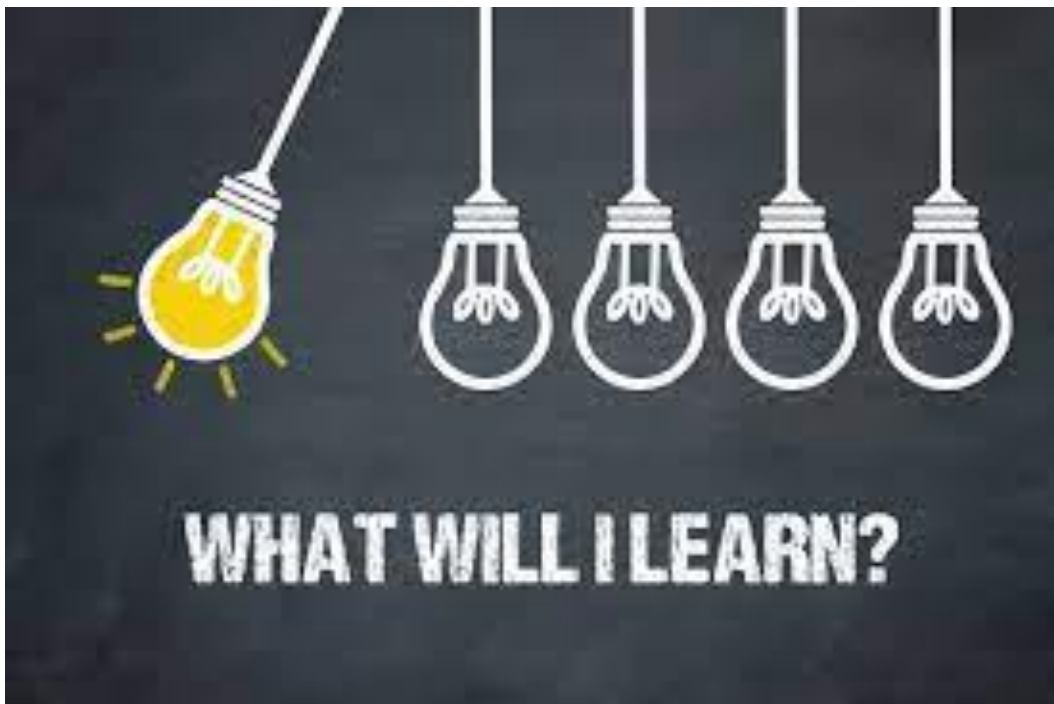
- **Scheduling Concepts:** Scheduling involves fine-tuning production plans at a specific, short-term level, utilizing
 - forward scheduling (earliest start) or backward scheduling (meeting a due date).
- **Loading and Scheduling Criteria:** Loading assigns jobs to workstations, with finite loading considering capacity. Key criteria include minimizing completion time, maximizing utilization, and minimizing work-in-process (WIP).
- **Sequencing Rules:** Practical rules for single-machine sequencing include First Come First Served (FCFS), Shortest Processing Time (SPT), Earliest Due Date (EDD), and Critical Ratio (CR).
- **Performance Metrics for Sequencing:** Rules are evaluated based on mean flow time, average tardiness, and utilization percentage.
- **Sequencing Theory and Algorithms:** Theorem 1 states that SPT minimizes mean flow time. Specific algorithms like Moore's minimize the number of tardy jobs, while Lawler's algorithm minimizes maximum lateness and handles precedence constraints.

- **Multiple Machine Scheduling:** Scheduling n jobs on m machines is complex; for two machines, the Johnson algorithm provides an optimal permutation schedule to minimize total flow time.
- **Assembly Line Balancing:** This involves grouping tasks into workstations based on a desired cycle time (C) to maximize efficiency and minimize idle time.
- **Project Management Phases:** Large outcomes are managed through three distinct phases: planning (defining goals and resources), scheduling (linking activities and people), and controlling (tracking progress and mitigating risk).
- **Work Breakdown Structure (WBS):** This is a necessary decomposition exercise that breaks projects into manageable parts, components, and finally specific work packages.
- **Project Scheduling Tools:** Common methods include Gantt charts for simple visualization and network-based methods like the Critical Path Method (CPM) and Project Evaluation and Review Technique (PERT) for analyzing interdependencies and randomness.

Agenda Course 5

- Inventory Planning for known demand
 - The role of inventory
 - Balance between different cost elements
 - Different inventory planning policies
 - EOQ sensitivity
- Inventory Planning for unknown demand
 - Stochastic demand
 - News vendor model
 - Multi-echelon systems and bullwhip

Learning Objectives



Understand the different types of inventory

Explain the role of inventory

Understand and use the EOQ model for independent demand

Compute a re-order point

Manage inventory for random demand

Why do we need inventory?

- If Supply = Demand & everything works perfectly

=> little need for inventory, goods are made at the same rate as the demand => line flow system!

- In real life:

- Product variability
- Demand variability
- Supply variability
- Batch production ...

=> Need to carry stock

Inventory Management

- Have a look at the income statement and balance sheet of below brand and calculate inventory over income
- What do you observe?
 - Schneider
 - VW
 - BMW
 - Renault
 - Nike
 - Zara
 - Apple
 - Amazon

CONDENSED CONSOLIDATED STATEMENTS OF OPERATIONS (Unaudited)

(In millions, except number of shares, which are reflected in thousands, and per-share amounts)

	Three Months Ended		Twelve Months Ended	
	September 28, 2024	September 30, 2023	September 28, 2024	September 30, 2023
Net sales:				
Products	\$ 69,958	\$ 67,184	\$ 294,866	\$ 298,085
Services	24,972	22,314	96,169	85,200
Total net sales ⁽¹⁾	94,930	89,498	391,035	383,285
			September 28, 2024	September 30, 2023

Inv/rev= 1,8%**ASSETS:**

Current assets:

Cash and cash equivalents	\$ 29,943	\$ 29,965
Marketable securities	35,228	31,590
Accounts receivable, net	33,410	29,508
Vendor non-trade receivables	32,833	31,477
Inventories	7,286	6,331
Other current assets	14,287	14,695
Total current assets	152,987	143,566

RESULTS OF OPERATIONS

(Dollars in millions, except per share data)

	FISCAL 2025	FISCAL 2024	% CHANGE	FISCAL 2023	% CHANGE
Revenues	\$ 46,309	\$ 51,362	-10%	\$ 51,217	0%
Cost of sales	26,519	28,475	-7%	28,925	-2%
Gross profit	19,790	22,887	-14%	22,292	3%

Inv/rev= 16%

NIKE, INC.

(In millions)

	MAY 31,	
	2025	2024
ASSETS		
Current assets:		
Cash and equivalents	\$ 7,464	\$ 9,860
Short-term investments	1,687	1,722
Accounts receivable, net	4,717	4,427
Inventories	7,489	7,519
Prepaid expenses and other current assets	2,005	1,854

Net Sales:

North America

International

AWS

Consolidated

Year Ended December 31,

2022

2023

\$ 315,880

\$ 352,828

118,007

131,200

80,096

90,757

\$ 513,983

\$ 574,785

Inv/rev= 6%

AMAZON.COM, INC.
CONSOLIDATED BALANCE SHEETS
(in millions, except per share data)

December 31,

2022

2023

ASSETS

Current assets:

Cash and cash equivalents

\$ 53,888

\$ 73,387

Marketable securities

16,138

13,393

Inventories

34,405

33,318

Accounts receivable, net and other

42,360

52,253

3.2.1. Compte de résultat consolidé

(en millions d'euros)

	Notes	2024	2023
Chiffre d'affaires	4	56 232	52 376

Renault Inv/rev= 10%

Actifs courants			
Stocks	14	5 468	4 924
Créances de Financement des ventes	15	54 355	49 615
Créances clients de l'Automobile	16	990	825

OTHER FINANCIAL PERFORMANCE FIGURES

in € million	2020	2021	2022	2023	2024	Change in %
Group revenues	98,990	111,239	142,610	155,498	142,380	- 8.4

BMW Inv/rev= 17%

NET ASSETS POSITION OF THE BMW GROUP

BMW Group Condensed Balance Sheet at 31 December

in € million	2024	2023	Change in %	Currency-adjusted change* in %	Proportion of balance sheet total in %
ASSETS					
Intangible assets	20,220	20,022	1.0	- 0.4	7.6
Property, plant and equipment	39,581	35,266	12.2	10.4	14.8
Leased products	48,838	43,118	13.3	11.2	18.2
Investments accounted for using the equity method	553	443	24.8	24.8	0.2
Other investments	1,099	1,197	- 8.2	- 12.0	0.4
Receivables from sales financing	93,718	87,355	7.3	4.7	35.0
Financial assets	3,399	5,518	- 38.4	- 39.1	1.3
Deferred and current tax	4,560	3,630	25.6	23.4	1.7
Other assets	9,256	9,133	1.3	0.8	3.5
Inventories	24,387	23,719	2.8	1.1	9.1

5.1 Compte de résultat consolidé

(en millions d'euros sauf le résultat par action)

	Note	2024	2023
Chiffre d'affaires	3	38 153	35 902

Schneider Inv/rev= 14%

ACTIFS COURANTS :

Stocks et en-cours	15	5 411	4 519
Clients et créances d'exploitation	16	9 364	8 388

ZARA

1. Key Financial Data (FY2025)

- **Total Revenue (Net Sales):** €39,864 million
- **Inventory Value (as of Jan 31, 2026):** €3,249 million

2. Calculation

To find the inventory value as a percentage of revenue:

$$\text{Ratio} = \left(\frac{\text{Inventory}}{\text{Revenue}} \right) \times 100$$

$$\text{Ratio} = \left(\frac{3,249}{39,864} \right) \times 100 \approx \mathbf{8.15\%}$$

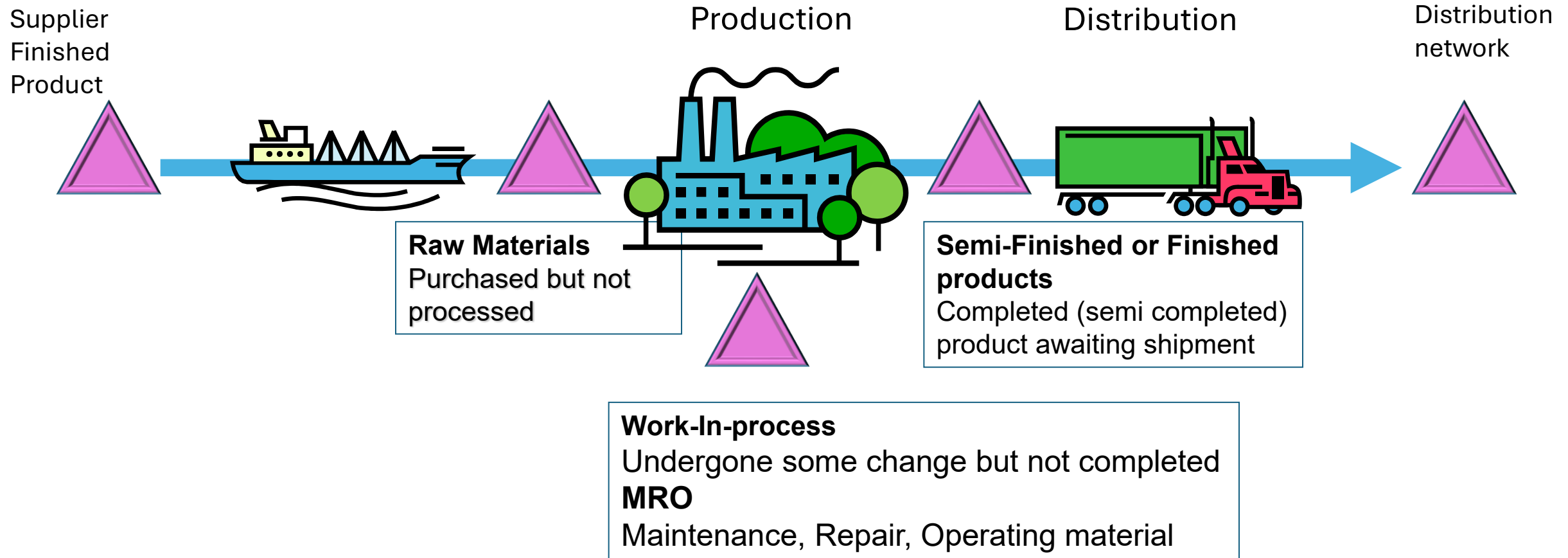
Inventory: different decisions to be made

- **Supply Chain Decisions** Strategic
 - What are the potential alternatives to inventory?
 - How should the product be designed?

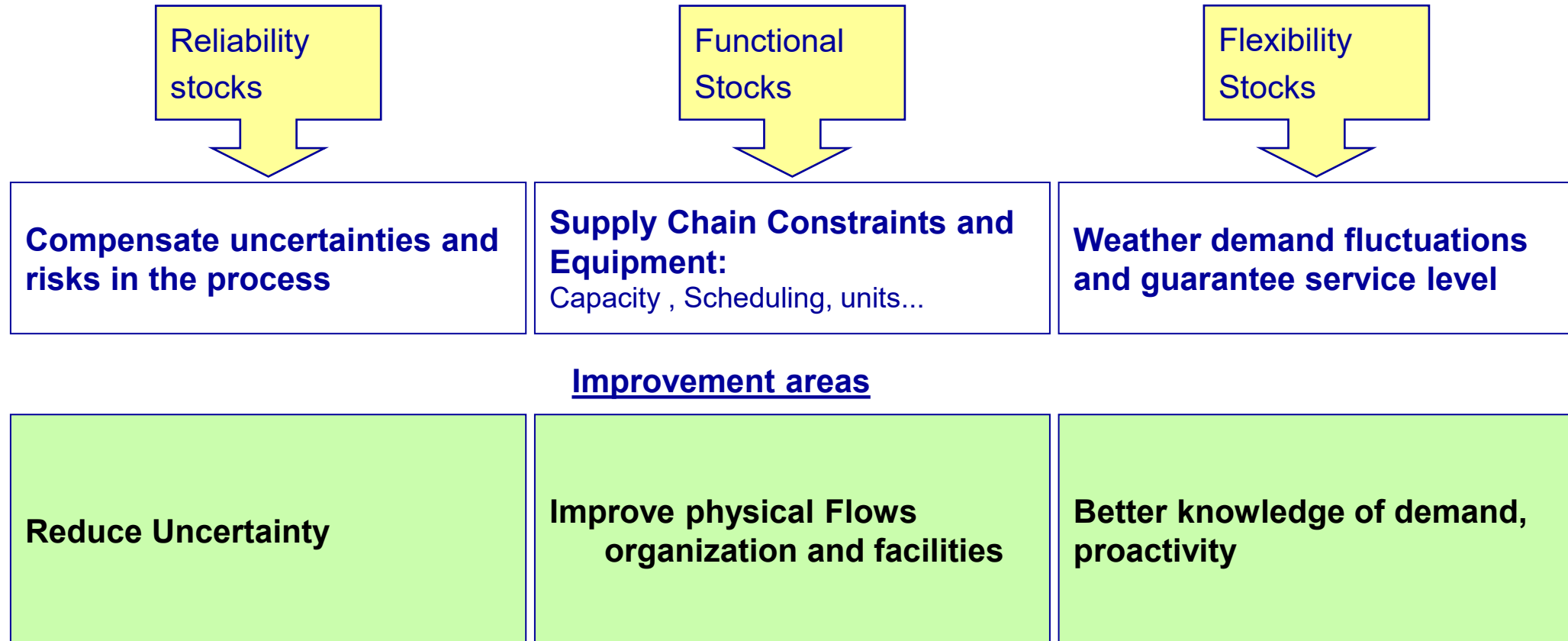
- **Deployment Decisions** Tactical
 - What items should be carried as inventory?
 - In what form should they be maintained?
 - How much of each should be held and where?

- **Replenishment Decisions** Operational
 - How often should inventory status be determined?
 - When should a replenishment decision be made?
 - How large should the replenishment be?

Different Types of inventory to play with



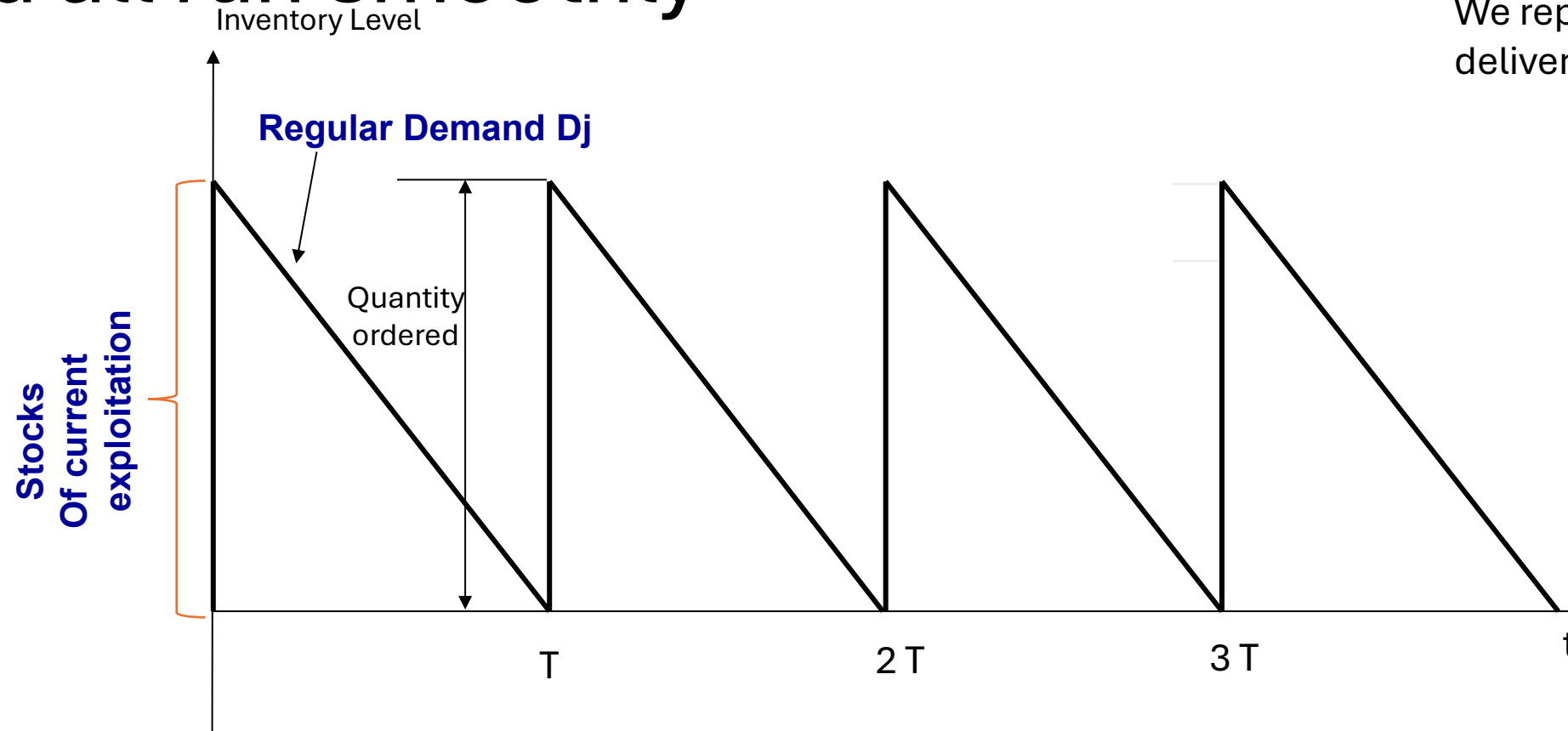
Inventory: different purposes



Inventory profile when demand is stable and all run smoothly



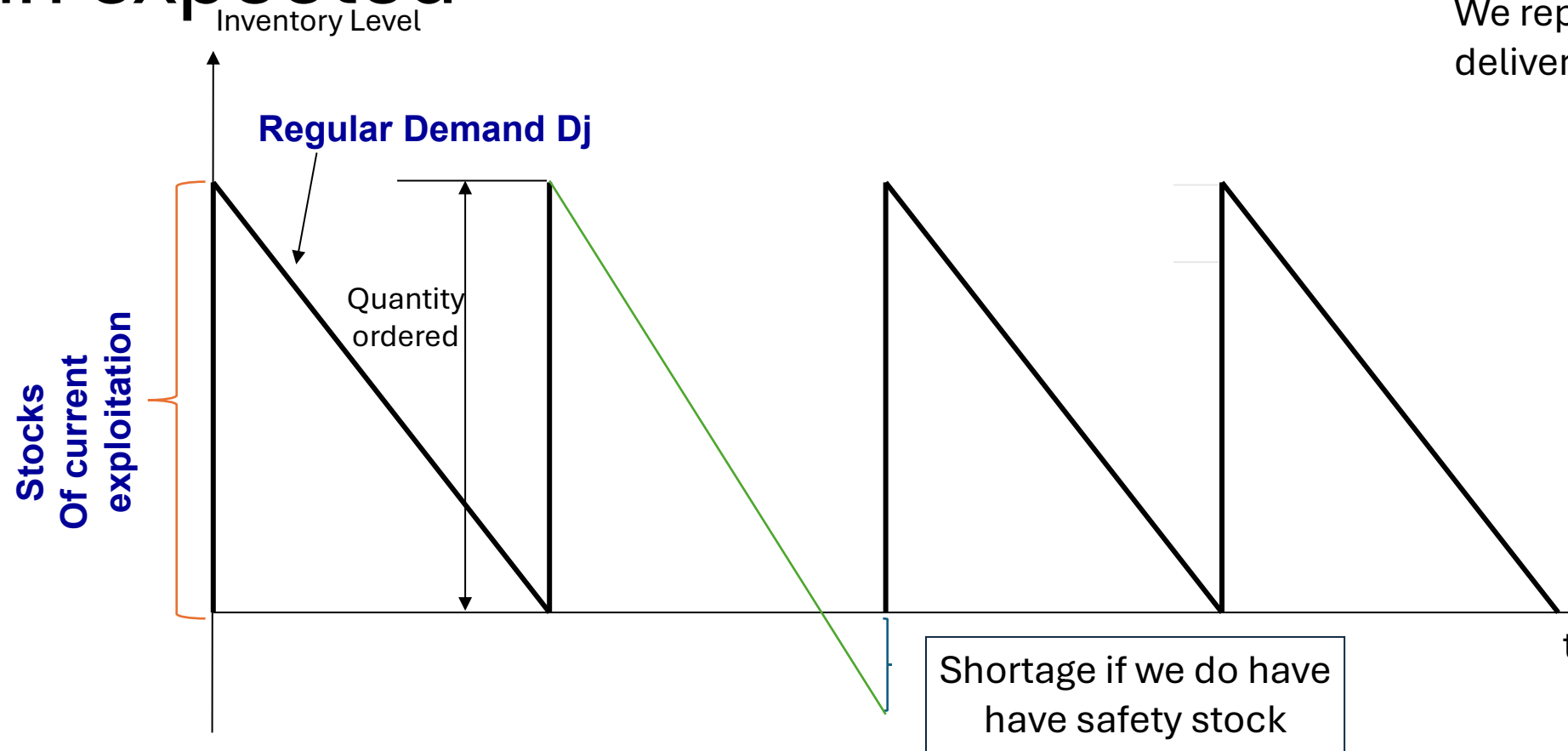
We replenish every T and get delivered immediately.



Inventory profile when demand is stronger than expected

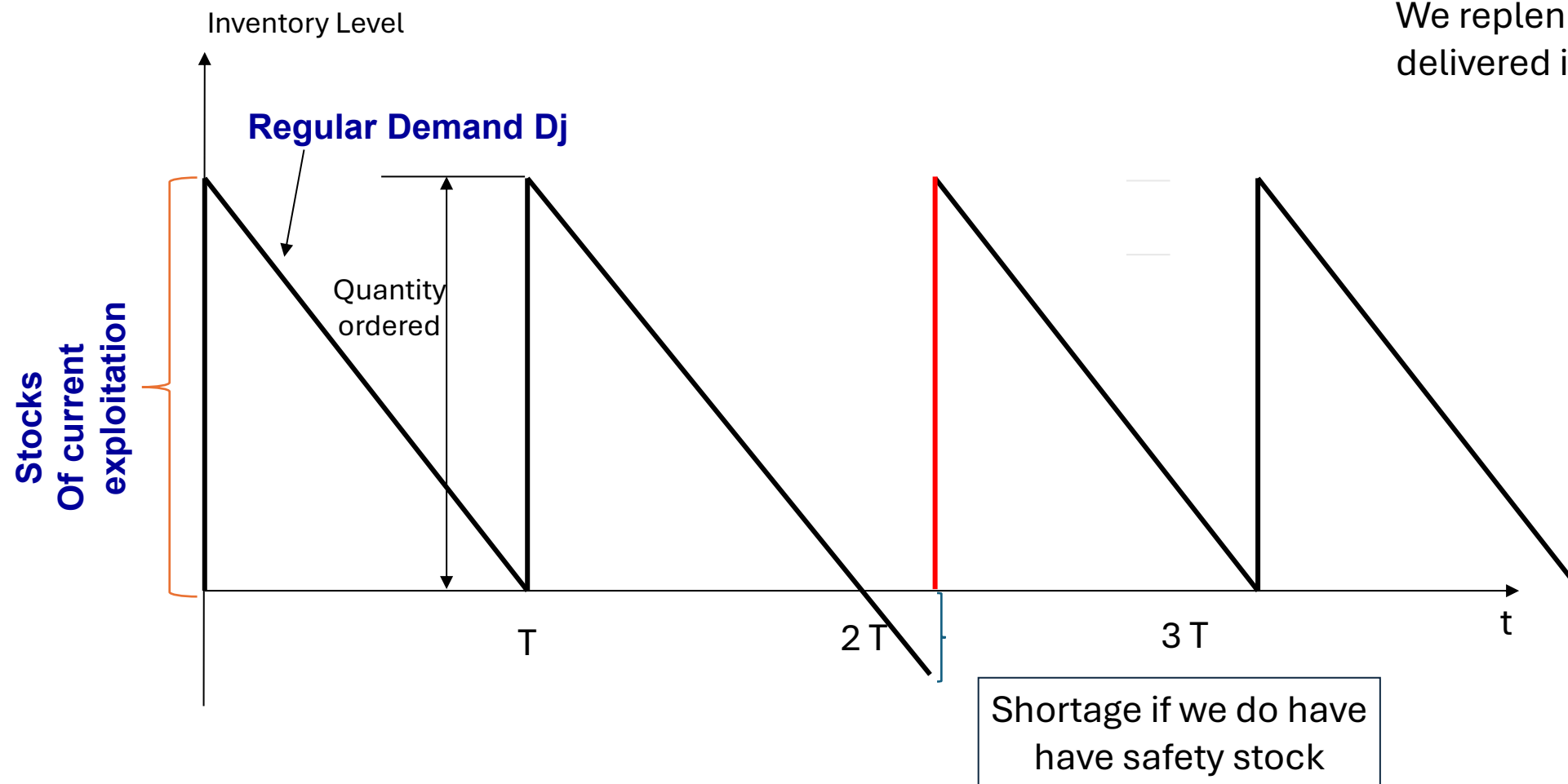


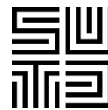
We replenish every T and get delivered immediately.



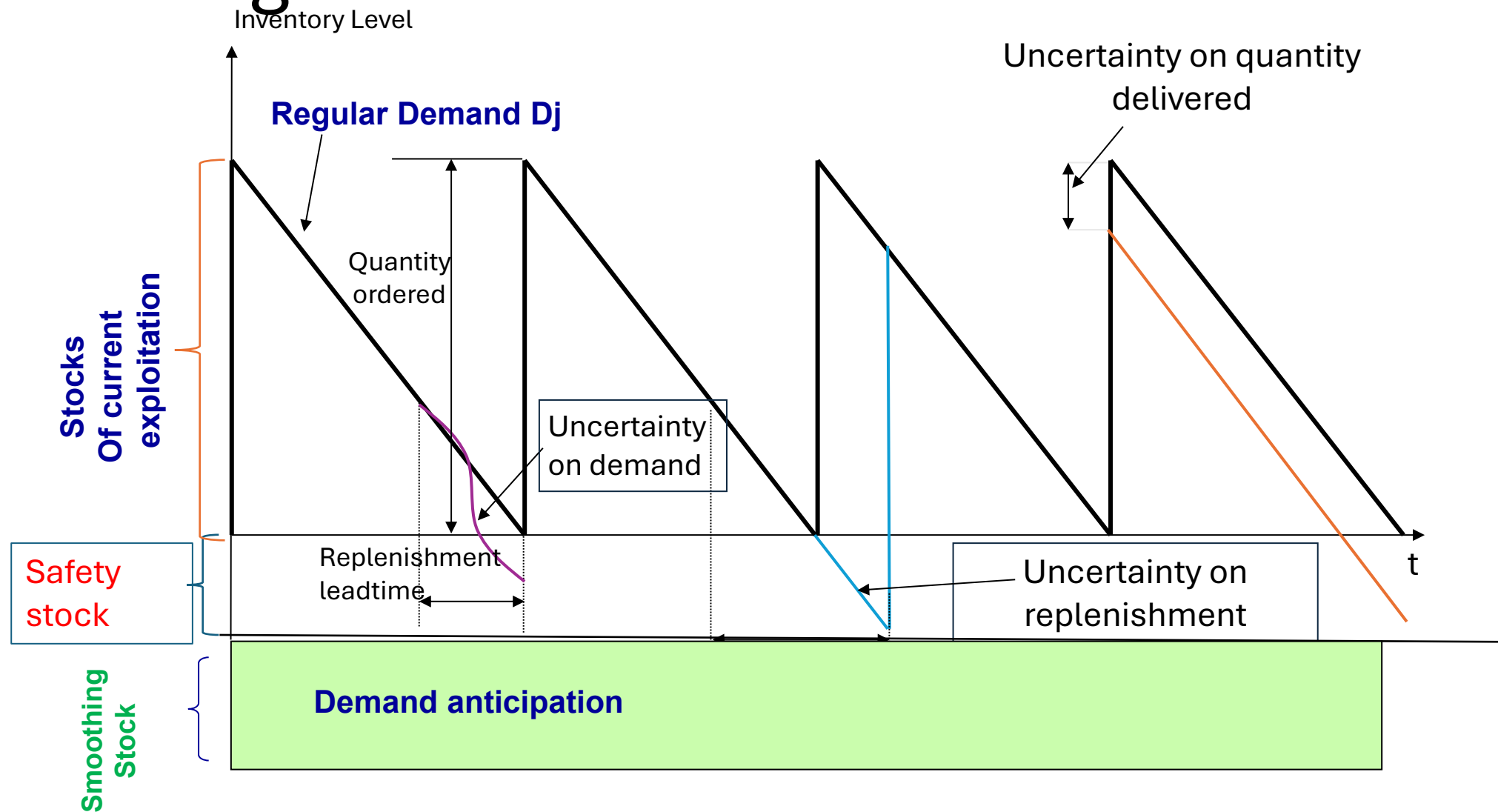
Inventory profile when supplier is late

We replenish every T and get delivered immediately.

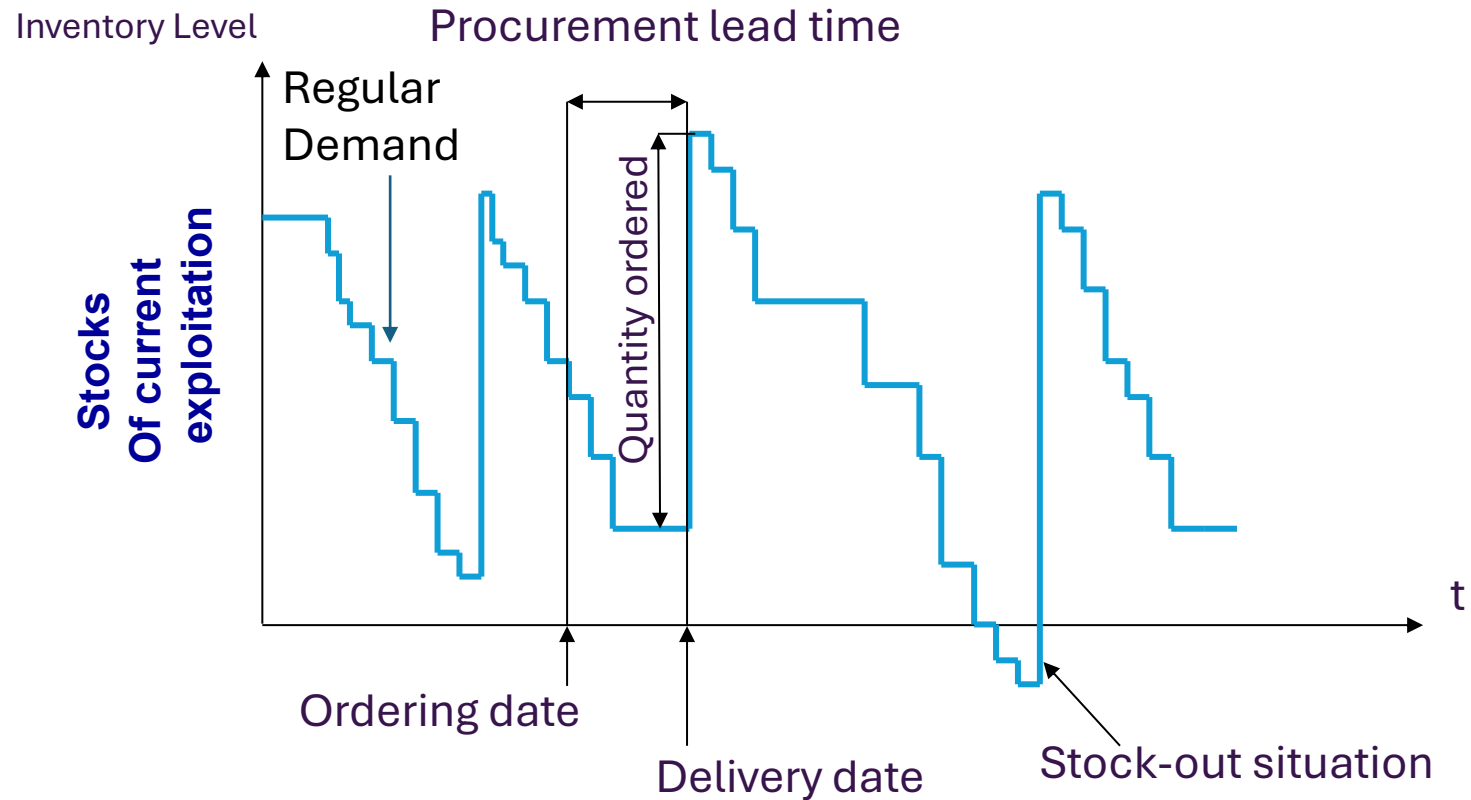




Inventory profile with safety stock and smoothing stock



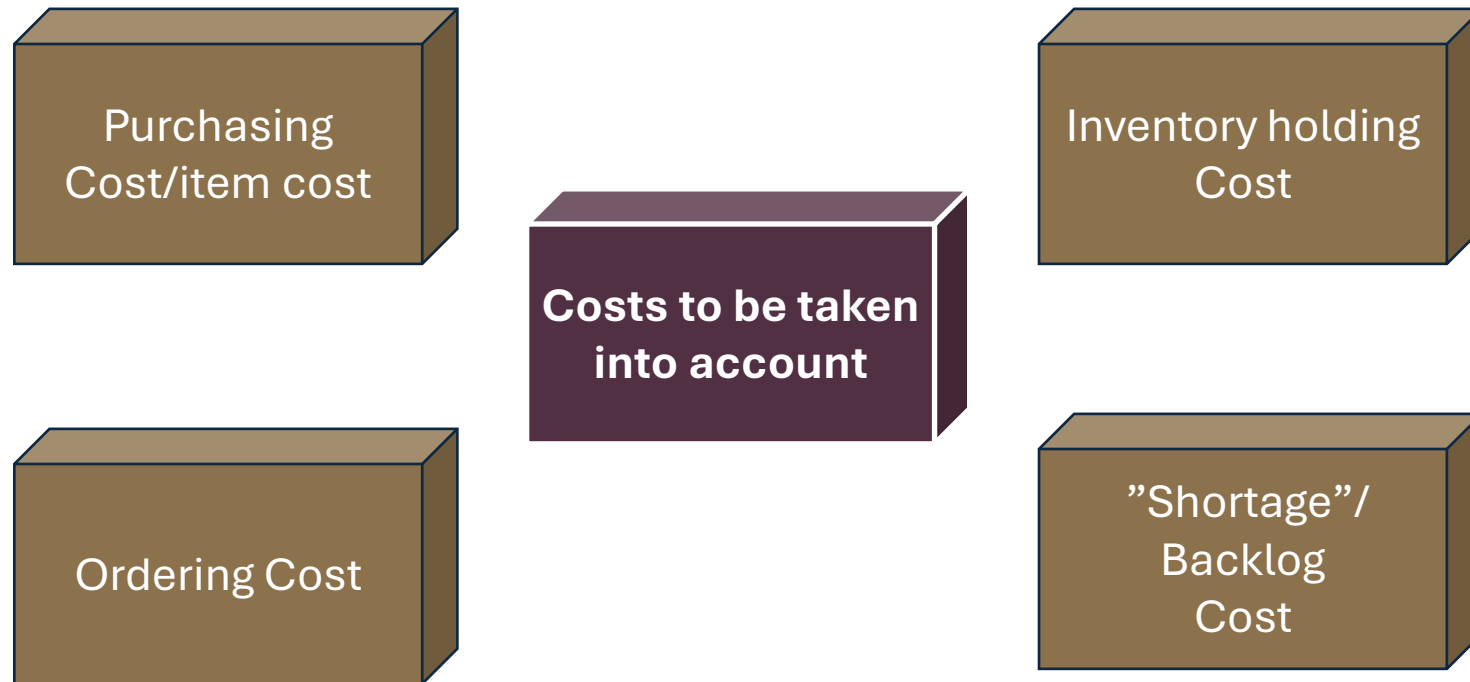
Inventory profile in reality



Inventory dependency to other variables

- Demand
 - Constant vs Variable
 - Known vs Random
 - Continuous vs Discrete
- Lead Time
 - Instantaneous
 - Constant vs Variable
 - Deterministic vs Stochastic
 - Internally Replenished
- Dependence of Items
 - Independent
 - Correlated
 - Indentured
- Review Time
 - Continuous vs Periodic
- Number of Locations
 - One vs Multi vs Multi-Echelon
- Capacity / Resources
 - Unlimited
 - Limited / Constrained
- Discounts
 - None
 - All Units vs Incremental vs One Time
- Excess Demand
 - None
 - All orders are backordered
 - Lost orders
 - Substitution
- Perishability
 - None
 - Uniform with time
 - Non-linear with time
- Planning Horizon
 - Single Period
 - Finite Period
 - Infinite
- Number of Items
 - One vs Many
- Form of Product
 - Single Stage
 - Multi-Stage

Costs attached to inventory



Procurement cost



Direct Purchasing costs: *Unit Purchase cost x Order size (the total amount of items ordered)*

- *Case 1: the unit cost does not depend on the order size*
- *Case 2: the unit cost depends on the order size, i.e., there is an incentive for purchasing the item in batches*

Ordering costs: *cost involved with procurement activities*

- *Different costs associated to an order:*
 - *the cost of preparing and placing orders*
 - *the cost of transportation of the items*
 - *the cost of handling and inspection of the shipment when it arrives*
- *Assumptions on the ordering costs*
 - *fixed costs (per order) + costs proportional to the order quantity*

Indirect
logistics
costs

=> Procurement cost = purchasing cost + ordering cost

Holding cost



Inventory holding costs: *Costs related to inventory management*

- *The financial resources*
- *The physical storage costs: building and handling equipment*
- *The Risk costs.*

=> Classical assumption: Inventory holding cost h :

- $h = r C$
- C : the unit product cost (€/product)
- r : the carrying charge (€ / time and invested €). it represents the cost of holding 1 euros of stock per year

Shortage cost



Shortage/Backlog costs: *when the customer service level requirements are not met*

lost sale case:

- *fixed cost each time a demand is lost*

backlogging case

- *fixed cost each time a demand is backlog*
- *the cost depends also on the time the customer has to wait*
- *Situation differs in B2B vs B2C and upon brand value*

We will ignore shortage cost in our model as it is difficult to estimate.

Main inventory models



Deterministic Models:

- Economic Order Quantity, EOQ
 - Economic Production Quantity, EPQ
 - EOQ with quantity discounts
 - Multi product EOQ
 - ...

EOQ extension:

- Countinous Review: (r, Q) policy
- Periodic Review: (T, S) policy

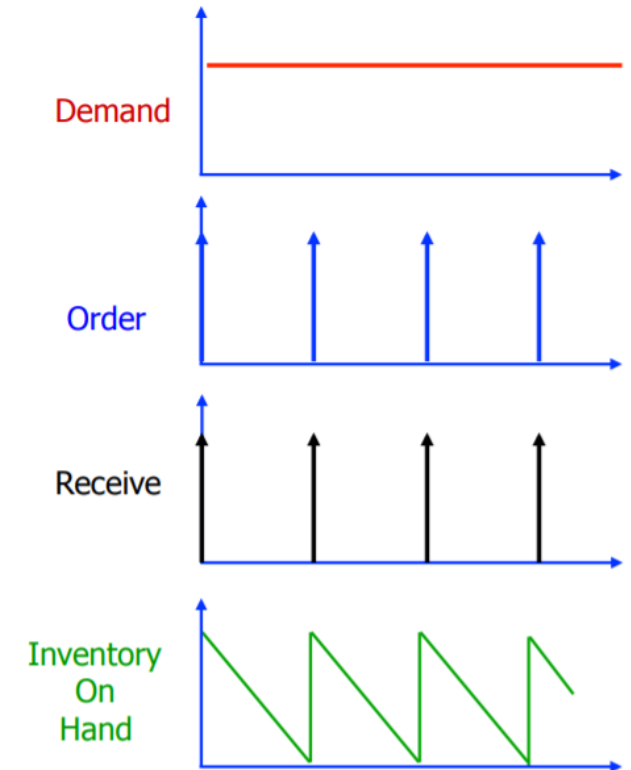
Single Order Model: next class focus

- Newsboy model

EOQ: variable assumptions



- Demand
 - **Constant** vs Variable
 - **Known** vs Random
 - **Continuous** vs Discrete
- Lead Time
 - **Instantaneous**
 - Constant vs Variable
 - Deterministic vs Stochastic
 - Internally Replenished
- Dependence of Items
 - **Independent**
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- Number of Items
 - **One** vs Many
- Form of Product
 - **Single Stage**
 - Multi-Stage



EOQ model: Wilson formula



C : unit purchase cost (Euros)

D : Demand (number of item per unit of time)

A : ordering cost (Euros per order)

h : unit carrying cost (Euros per unit per unit of time; $h = \text{holding rate} \times C$)

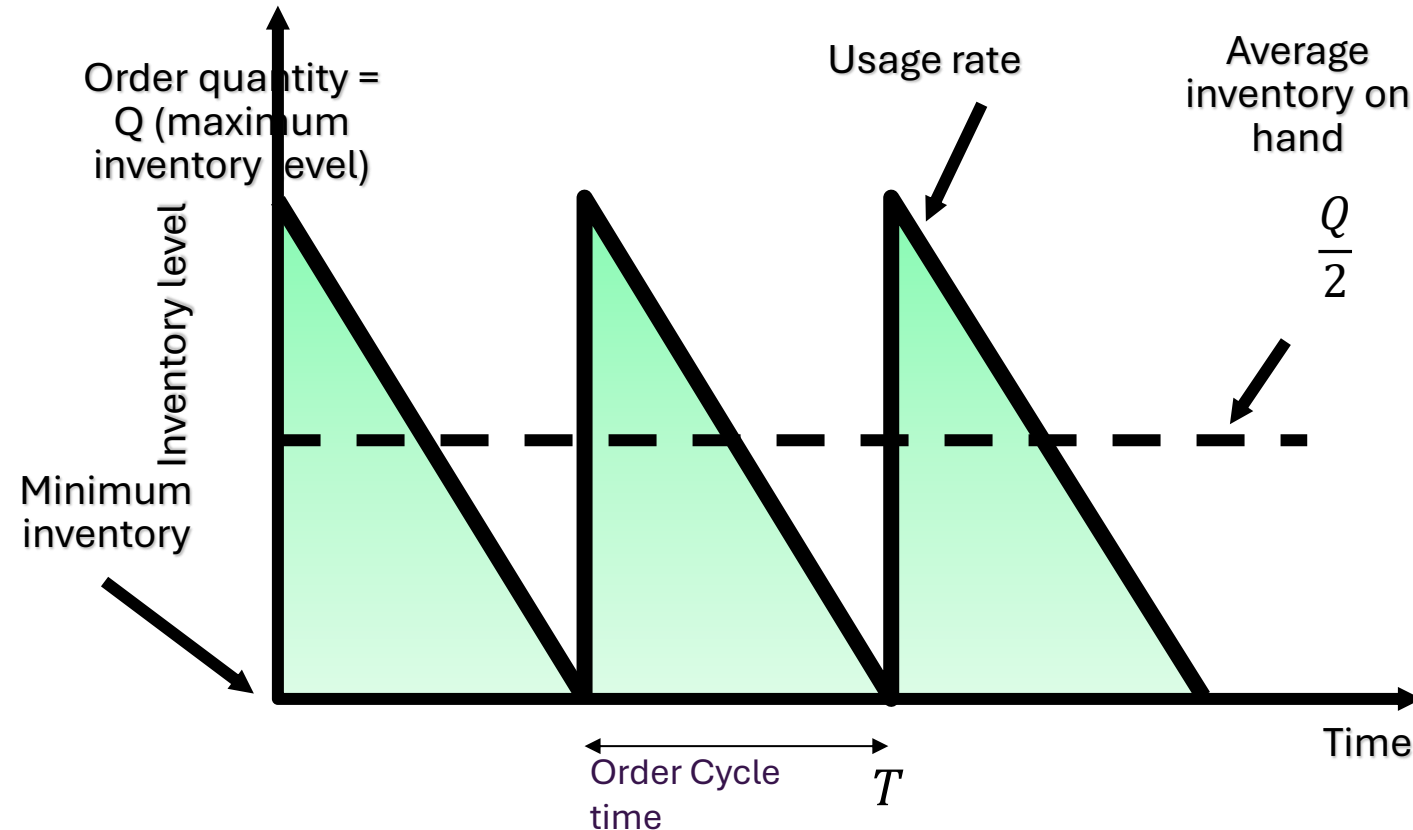
Q : ordering quantity (number of items per order)

T : cycle time (time between two successive orders)

N : ordering frequency (number of orders per time unit; $N = 1/T$)

$x(t)$: number of item in the stock at time t

$\bar{x}$: average inventory level



EOQ model: Wilson formula

How much to order each time to minimise the inventory cost?

Total ordering cost

- Purchase cost: $= C * D$
- Order cost : $= A \frac{D}{Q}$
- Total ordering cost : $= C * D + A \frac{D}{Q}$

Holding cost:

- Average stock: $\bar{x} = \frac{1}{T} \int_{t=0}^T x(t) dt = \frac{Q}{2}$ $h\bar{x} = h \frac{Q}{2}$
- Expected holding cost :

Expected total cost:

$$TC(Q) = CD + h \frac{Q}{2} + A \frac{D}{Q}$$

Expected variable cost

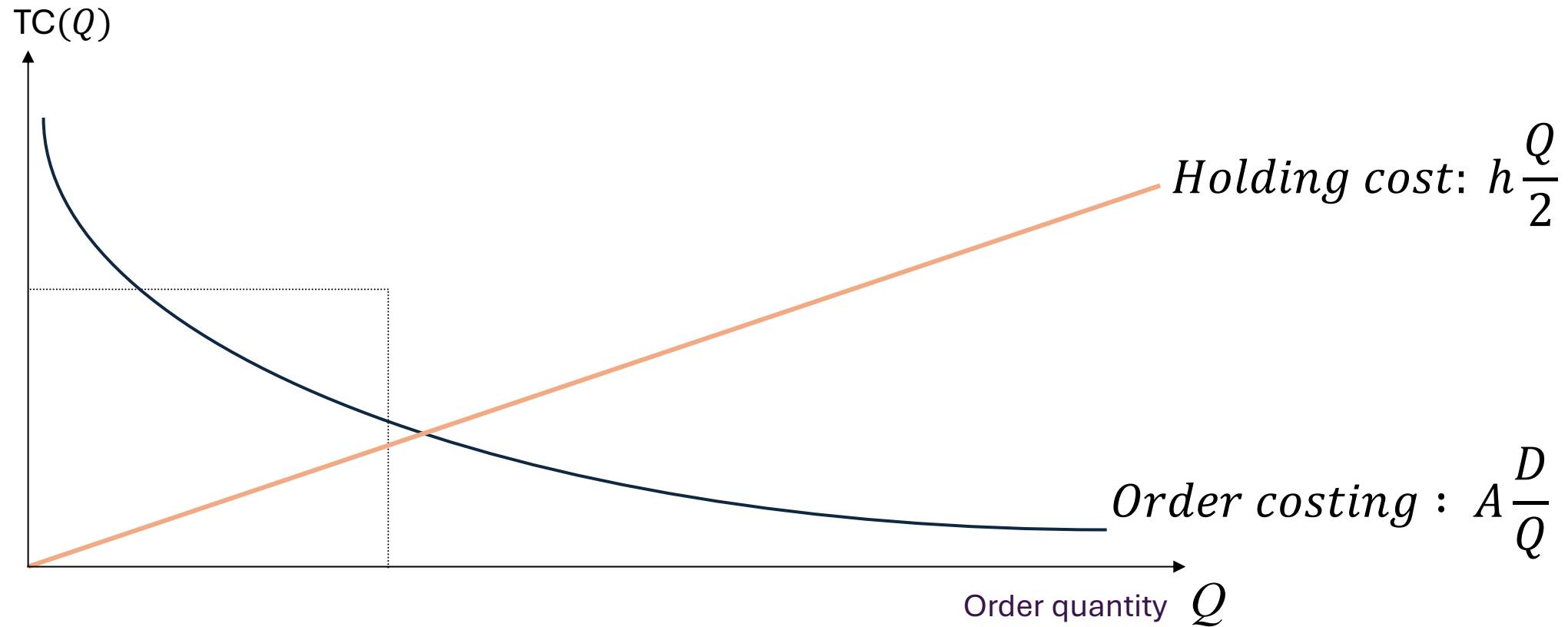
$$TVC(Q) = h \frac{Q}{2} + A \frac{D}{Q}$$

Fundamentals of Operations Management

Wilson Formula: EOQ Model Notations

- C : unit purchase cost (Euros)
- D : Demand (number of item per unit of time)
- A : ordering cost (Euros)
- h : unit carrying cost (Euros per unit per unit of time)
- Q : ordering quantity (number of items)
- T : cycle time (time between two successive orders)
- N : ordering frequency (number of orders per time unit)
- x(t) : number of item in the stock at time t
- $\bar{x}$: average inventory level

EOQ model: Wilson formula: 2 competing costs



EOQ model: example



D = 10 000 annual demand
C = 10 € unit purchase price
r = 20% holding charge =>
A = 100 € ordering cost

$$h = C \cdot r = 2 \text{ €uros}$$

Q	# order	Ordering cost (A)	Average Inventory	Holding cost	Variable cost
10000	1	100	5000	10000	10100

Variable cost:
 $100 + 10\,000 = 10\,100 \text{ €}$

We can order
10 000

1 order

Ordering cost
 $1 \cdot 100 = 100 \text{ €uros}$

Average inventory
($Q/2$ cf p45)
 $10\,000/2 = 5000$

Cost to hold inventory
 $5\,000 \cdot 2 = 10\,000 \text{ €uros}$

EOQ model: example



D = 10 000 annual demand
C = 10 € unit purchase price
r = 20% holding charge => **h = C*r = 2 €uros**
A = 100 € ordering cost

Q	# order	Ordering cost	Average Inventory	Holding cost	Variable cost
10000	1	100	5000	10000	10100
5000	2	200	2500	5000	5200

We can order
5 000

Variable cost
 $200 + 5\,000 = 5\,200 \text{ €}$



EOQ model: example

$D = 10\,000$ annual demand
 $C = 10\text{ €}$ unit purchase price
 $r = 20\%$ carrying charge $\Rightarrow$ $h = C \cdot r = 2\text{ €uros}$
 $A = 100\text{ €}$ ordering cost
 $Q^* = 1000$

Q	# order	Ordering cost	Average Inventory	Holding cost	Variable cost
10000	1	100	5000	10000	10100
5000	2	200	2500	5000	5200
2000	5	500	1000	2000	2500
1000	10	1000	500	1000	2000
500	20	2000	250	500	2500
100	100	10000	50	100	10100

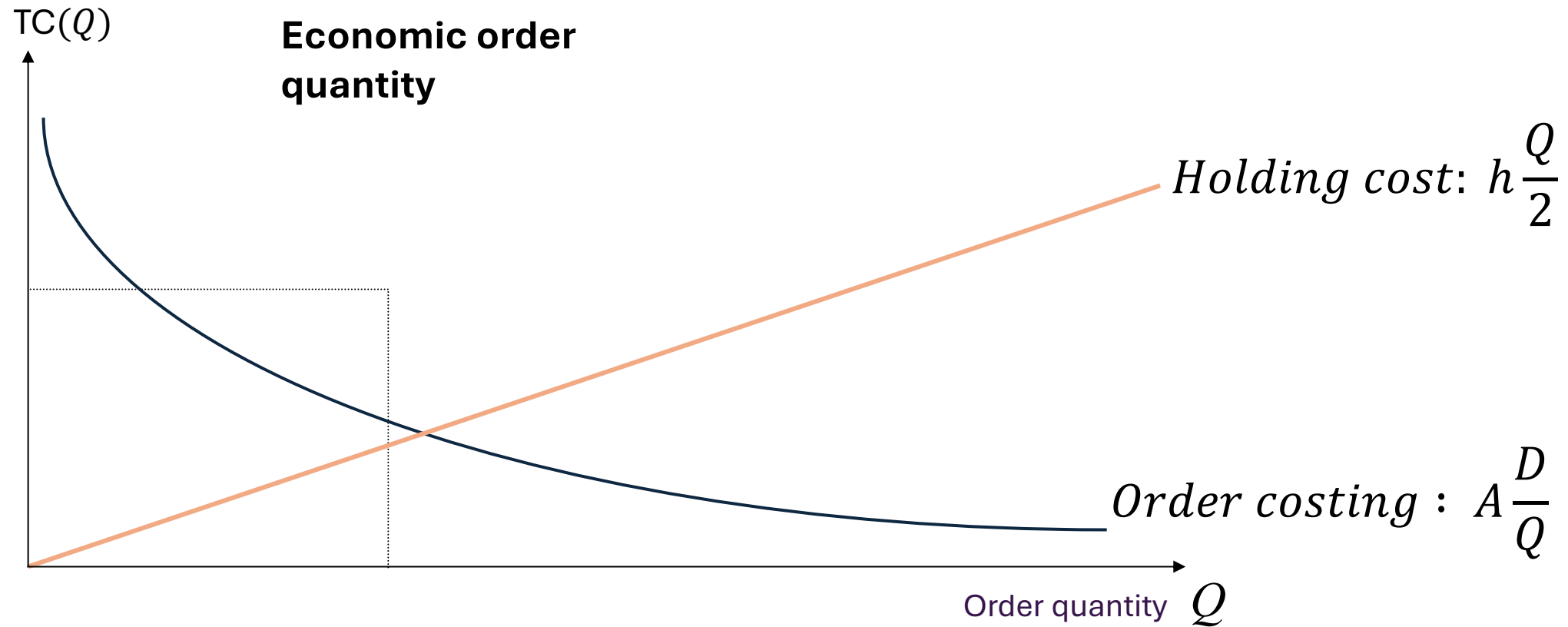


The variable cost decreases when the quantities ordered are reduced
Then it increases again. The optimum Ordered Q is Q^* .

EOQ model: Wilson formula



Minimise $TC(Q)$ $TC(Q) = CD + h\frac{Q}{2} + A\frac{D}{Q}$

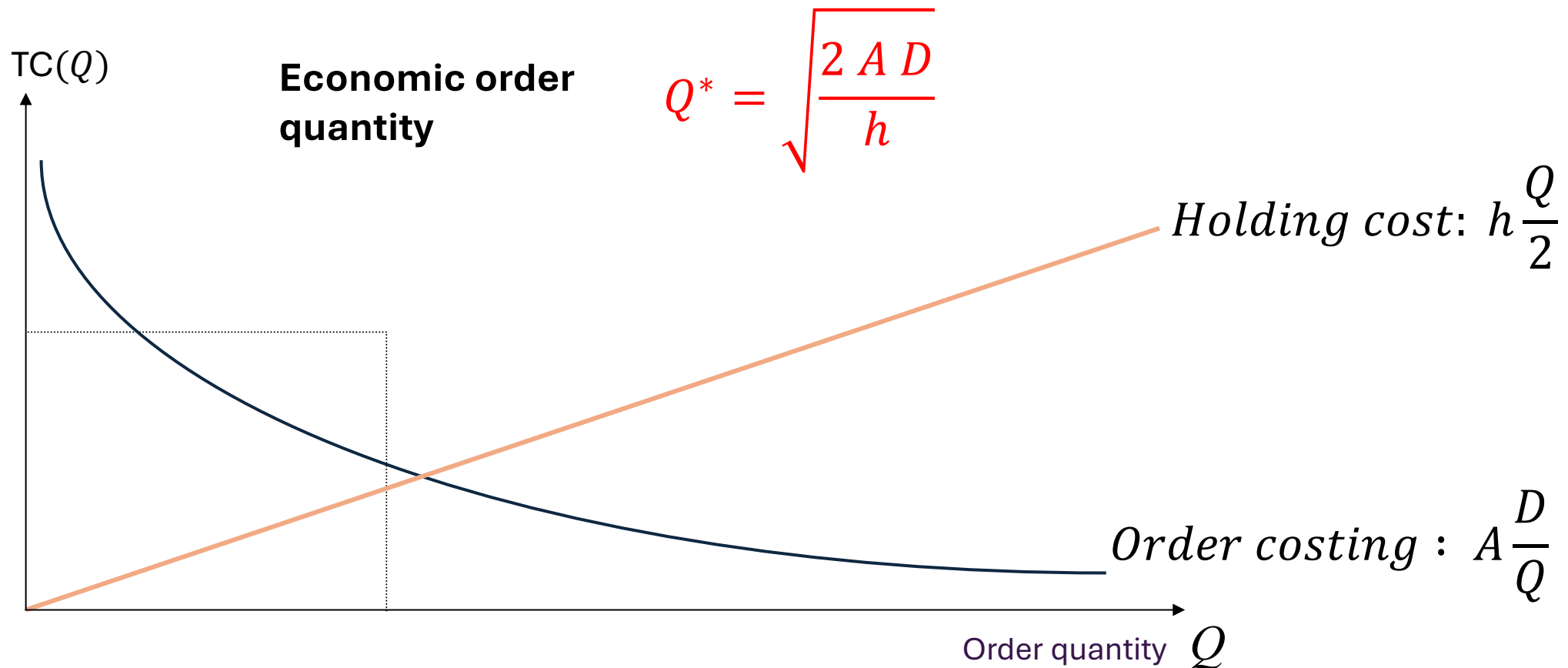


EOQ model: Wilson formula

Minimise $TC(Q) \Rightarrow \text{derivative } TC(Q)=0$

$$TC(Q) = CD + h \frac{Q}{2} + A \frac{D}{Q}$$

$$\frac{d(TC(Q))}{dQ} = d\left(A \frac{D}{Q} + h \frac{Q}{2}\right)/dQ = 0$$



EOQ model



- **Optimal** Total cost: $TC(Q^*) = CD + \sqrt{2hAD}$

Recall !!!

$$\blacktriangleright TC(Q) = CD + h \frac{Q}{2} + A \frac{D}{Q}$$

Optimal order quantity

$$Q^* = \sqrt{\frac{2AD}{h}}$$

- **Optimal** Variable Cost (dependant on Q) : $TVC(Q^*) = \sqrt{2hAD}$

- **Optimal** cycle time to place orders:

$$T^* = \frac{Q^*}{D} = \sqrt{\frac{2A}{hD}}$$

- **Optimal** ordering Frequency :

$$N^* = \frac{1}{T^*} = \sqrt{\frac{hD}{2A}}$$

EOQ model: exercise 1



The Papidou company uses 25,000 pinion assemblies per year and buys them for €5 each.

A: The ordering cost is €90.

h: The holding cost of an assembly is 1.4 €/year.

What is the most economical order quantity?

How many orders do you need to place per year?

What is the holding rate?

What is the overall cost for this item ? and then what is the actual cost price per item?

You noticed that your variable cost was higher than the theory. You spent 2900 euro. Which quantity did the team ordered instead of EOQ?

Wilson Formula: EOQ Model Notations

C : unit purchase cost (Euros)

D : Demand (number of item per unit of time)

A : ordering cost (Euros)

h : unit carrying cost (Euros per unit per unit of time)

Q : ordering quantity (number of items)

T : cycle time (time between two successive orders)

N : ordering frequency (number of orders per time unit)

$x(t)$: number of item in the stock at time t

$\bar{x}$: average inventory level

Inventory policies



Policies combination	Fixed Date	Variable Date
Fixed Quantity	Automatic replenishment	Reorder point method
Variable Quantity	Periodic inventory replenishment	Opportunistic Inventory buys

Deliveries are made on a fixed date
Delivered quantities are the same

Policies combination	Fixed Date	Variable Date
Fixed Quantity	Automatic replenishment	Reorder point method
Variable Quantity	Periodic inventory replenishment	Opportunistic Inventory buys

Applied to products with :

- Regular consumption
- Low value
- Class C

Inventory policies



Policies combination	Fixed Date	Variable Date
Fixed Quantity	Automatic replenishment	Reorder point method
Variable Quantity	Periodic inventory replenishment	Opportunistic Inventory buys

We define the level of stock that triggers the purchase order, so as to be delivered just as the last part is used.

Politics combination	Fixed Date	Variable Date
Fixed Quantity	Automatic Replenishment	Reorder point method
Variable Quantity	Periodic inventory replenishment	Opportunistic Inventory buys

This method is mainly used Class A products as they require a permanent monitoring of stocks resulting in a high management cost. Conversely, it makes it possible to avoid disruption on strategic parts.

Re-order point method:

The delivery of the goods is NOT instantaneous

Define the stock level that triggers the purchase order, so that it is delivered just when the last non-safety stock part is used.

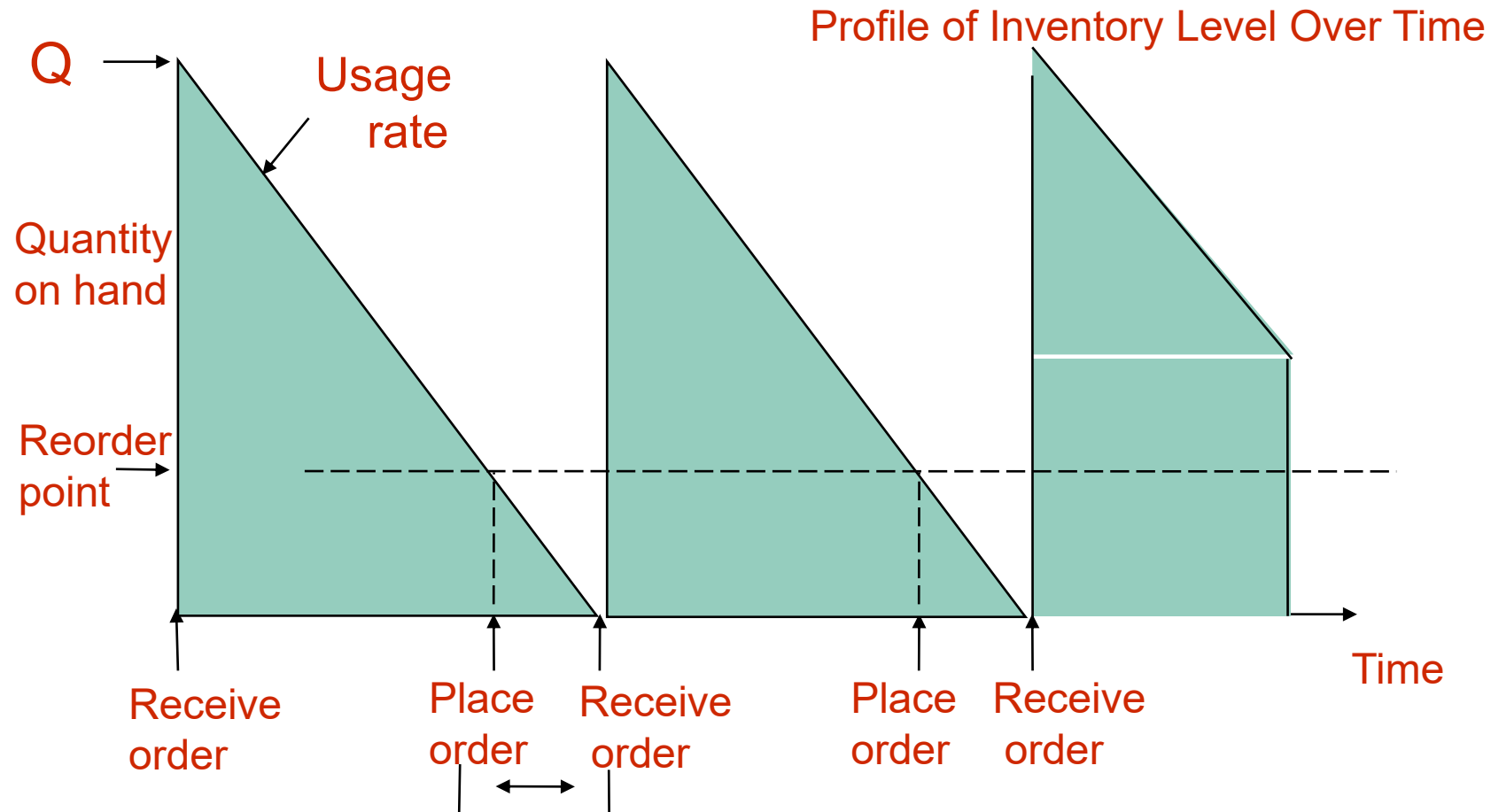
This stock level (order point) must allow needs to be met during the period from the order date to the delivery date.

Reorder point is also called reorder threshold.

Example: When the fuel level indicator in the car tank drops below $\frac{1}{4}$, I decide to refuel.
When I find a station I put in for 50€ of gasoline

Re-order point method:

The delivery of the goods is NOT instantaneous



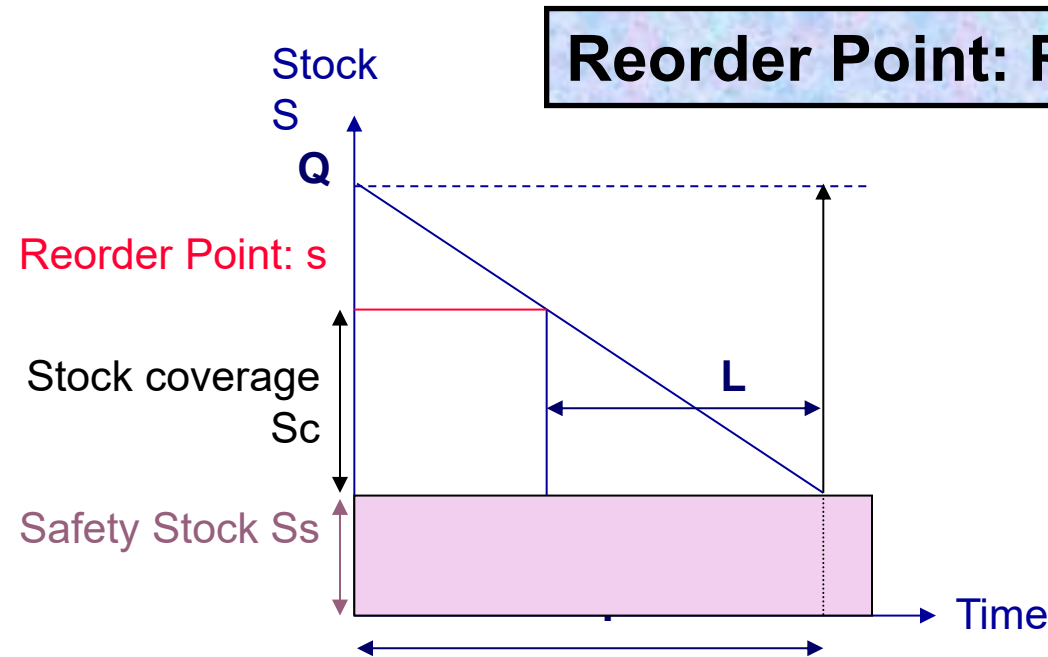
Re-order point method:

The delivery of the goods is NOT instantaneous

As there are uncertainties, a safety stock S_s is maintained.

The reorder point is equal to the cover stock + the safety stock.

The safety stock is determined by the inventory planner.



$$\text{Reorder Point: } ROP = S_c + S_s = (Q/T) * L + S_s$$

$$ROP = D_j * L + S_s$$

- Q order quantity after delivery
- T inventory consumption period
- L Delivery lead time (and / or production)



Re-order point method:

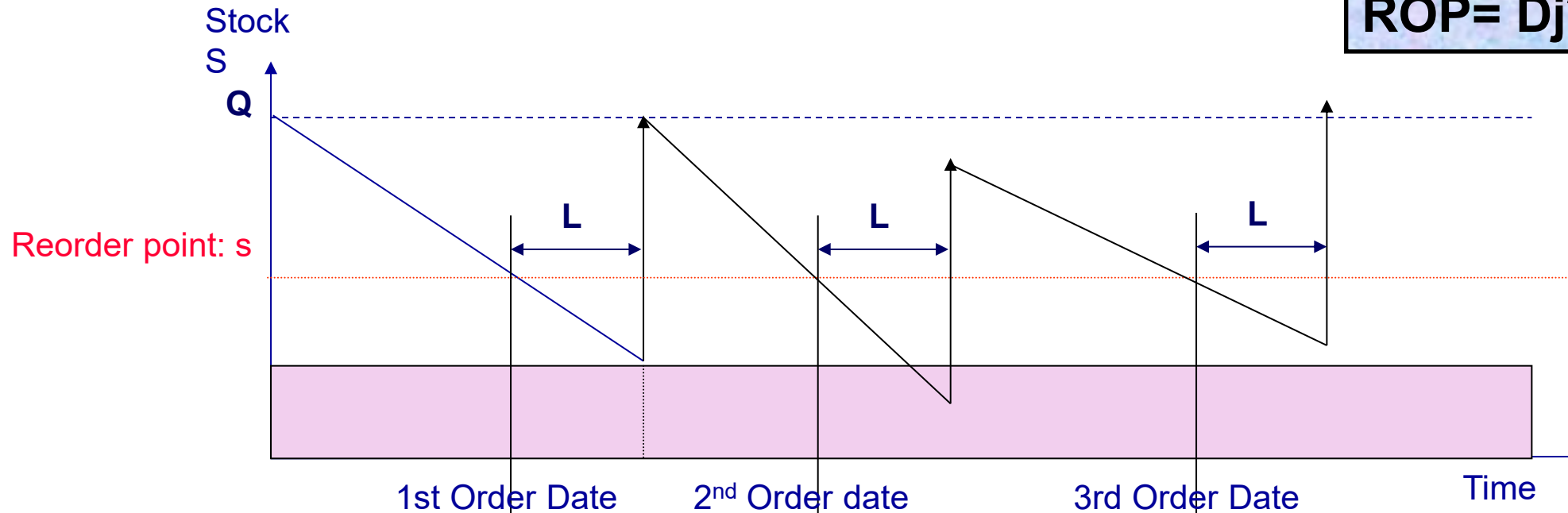
The delivery of the goods is NOT instantaneous

As consumption can vary over time, the order frequency is not fixed.

As the order quantity is fixed, the stock level after receipt is variable

$$\text{Reorder point : } ROP = S_c + S_s = (Q/T)*L + S_s$$

$$ROP = D_j * L + S_s$$



The order date is variable, it is the date when $S < S_c$

Re-order point method: exercise

The PCD company works 12 months a year. The number of working days may vary from month to month due to different parameters: public holidays, number of weekends, total number of days in the month, etc. It will be considered that there are on average 20 working days per month and that this number of days worked per month varies between 18 and 22.

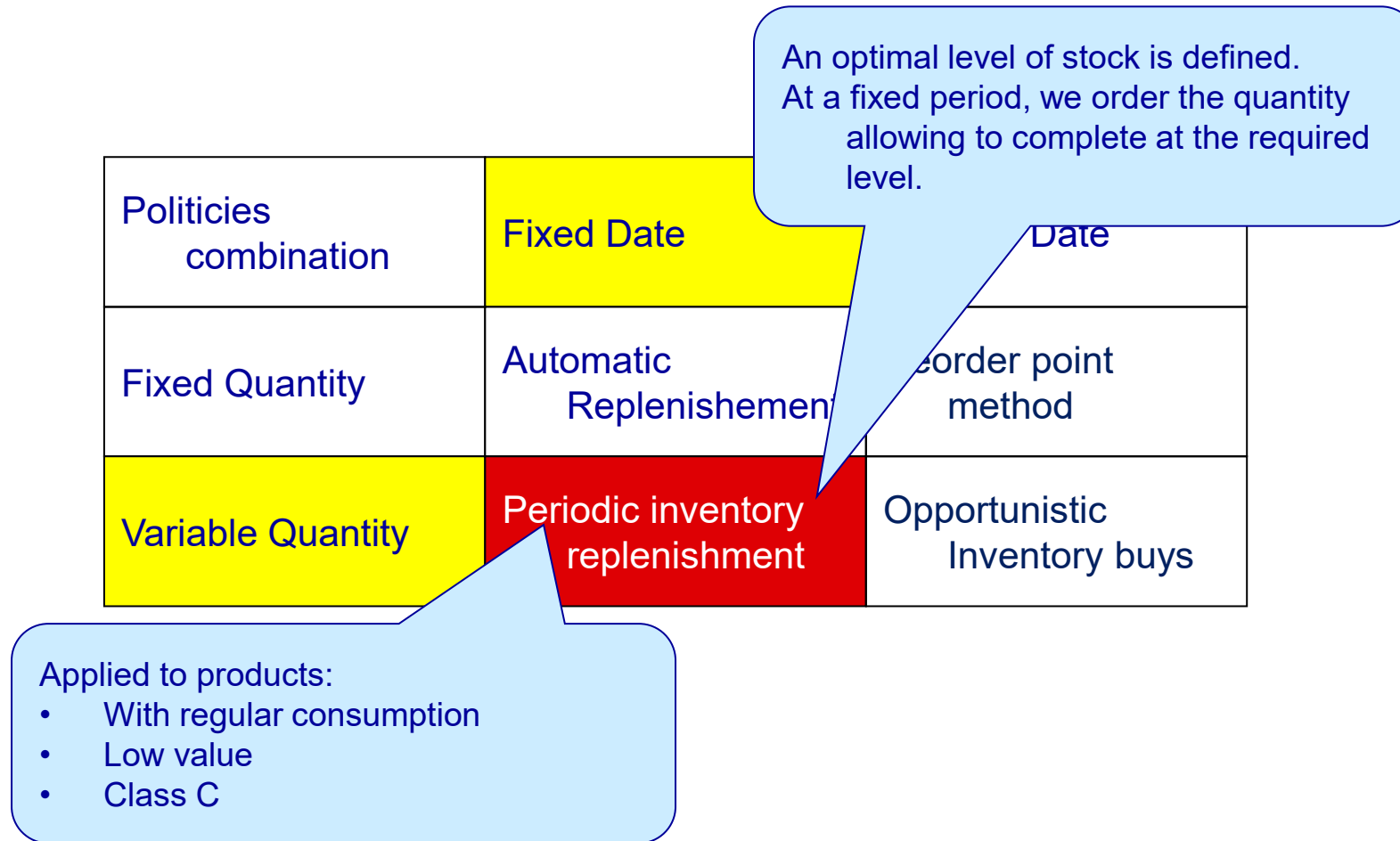
The QUECO item is replenished using the reorder point method.

The consumption of this item is 10,000 per year. The daily consumption can vary by more or less 10% compared to its average value. The unit purchase price of QUECO is 50 €. The cost of an order is 80 € and the annual holding ratio is 20%.

The average lead time from the supplier is 1 week (5 days), but it can come 2 days late.

- Calculate reorder point for the QUECO item.
- How much will you order from QUECO in each order?
- When will you place your next order?

Periodic inventory replenishment



Periodic replenishment method:



We place orders on a fixed date (every 10 days or the 1st of each month, etc.)

The quantity ordered is not fixed, it depends on the quantity remaining in stock

The quantity ordered makes it possible to replenish the stock at a level which will have been previously fixed.

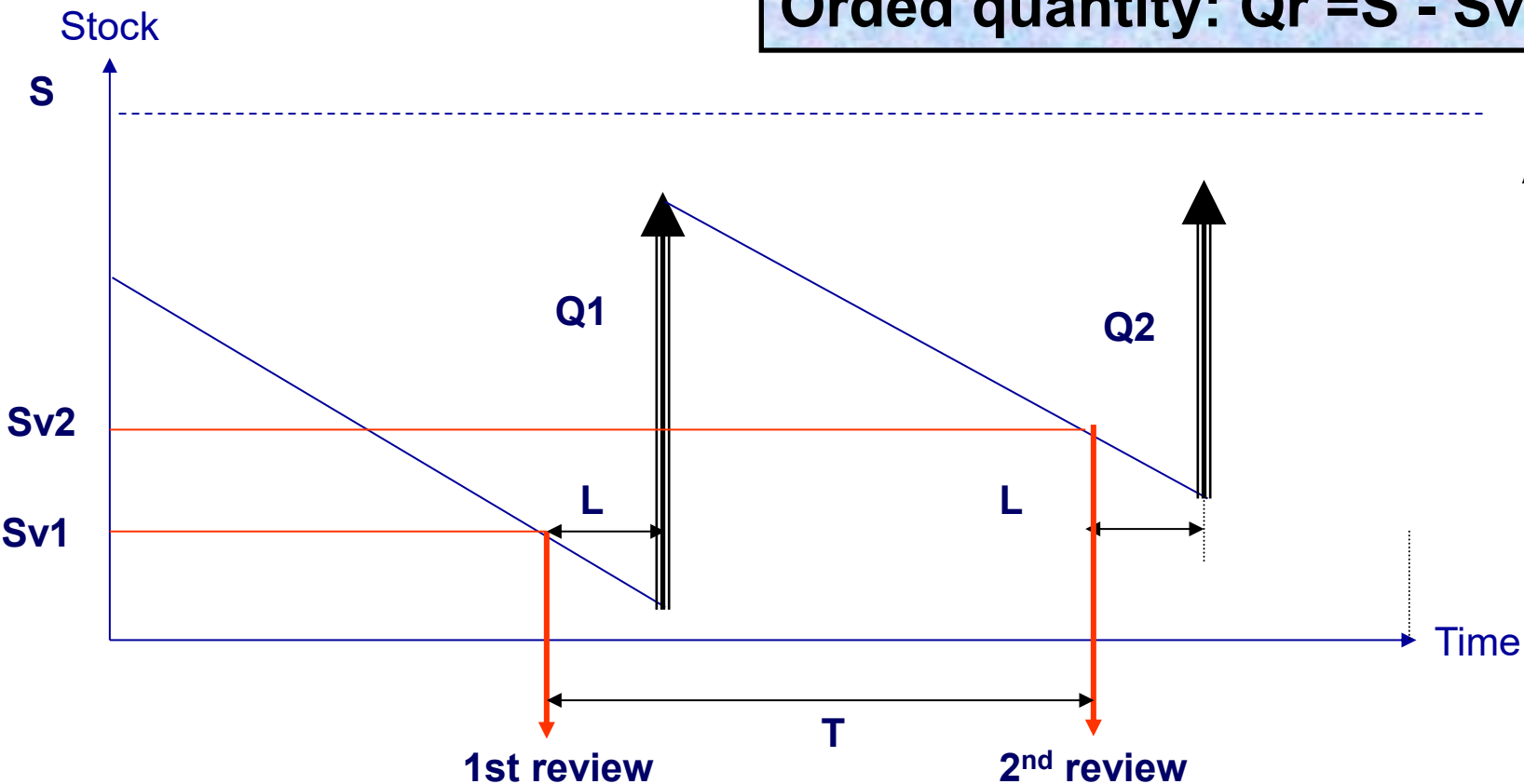
Example: Every Saturday morning, I go to fill up the gas tank of my car.
When I'm at the station I fill up the tank.

Periodic replenishment method:



Review dates are fixed : Every T
Variable ordered quantity: $Q1 \neq Q2$

Replenishment level $S = D_j^*(T+L) + S_s = Q^* + D_j^*L + S_s$
Ordered quantity: $Q_r = S - S_v$



As the number of days between 2 reviews might vary we approximate $D_j T \approx Q^*$

Periodic replenishment method: exercise



The Recomp company works 12 months per year ($18 < \text{average} = 20 < 22$ days/month). The consumption of this item is 100,000 per year, it is very regular.

The unit purchase price of DéFix is 1.16 €. The cost of an order is 100 Euros and the annual holding ratio is 25%.

The DéFix item is replenished following a periodic replenishment method: The review process takes place on the first working day of each month which allows to order the quantity needed. As the supplier lead time (10 days) is not very reliable, we chose a safety stock of 2000 units. We are on the first working day of the month, thus a review day.

DéFix's stock inventory indicates that there are 5750 items left.

How many items do we order ?

How many days included the previous month ?

What is the maximum number of days the supplier can be late in delivering this order before we run out?

EOQ model sensitivity

- Sensitivity to order quantity
- Sensitivity to demand
- Sensitivity to ordering time

EOQ model sensitivity to Q



$$Z(Q^*) = \sqrt{2 * A * h * D}$$

$$Z(Q) = h \frac{Q}{2} + A \frac{D}{Q}$$

$$Q^* = \sqrt{\frac{2 * D * A}{h}}$$

$$Z(Q)/Z(Q^*) = h * Q / 2 \sqrt{2 * h * A * D} + 1/Q * (A * D / \sqrt{2 * h * A * D})$$

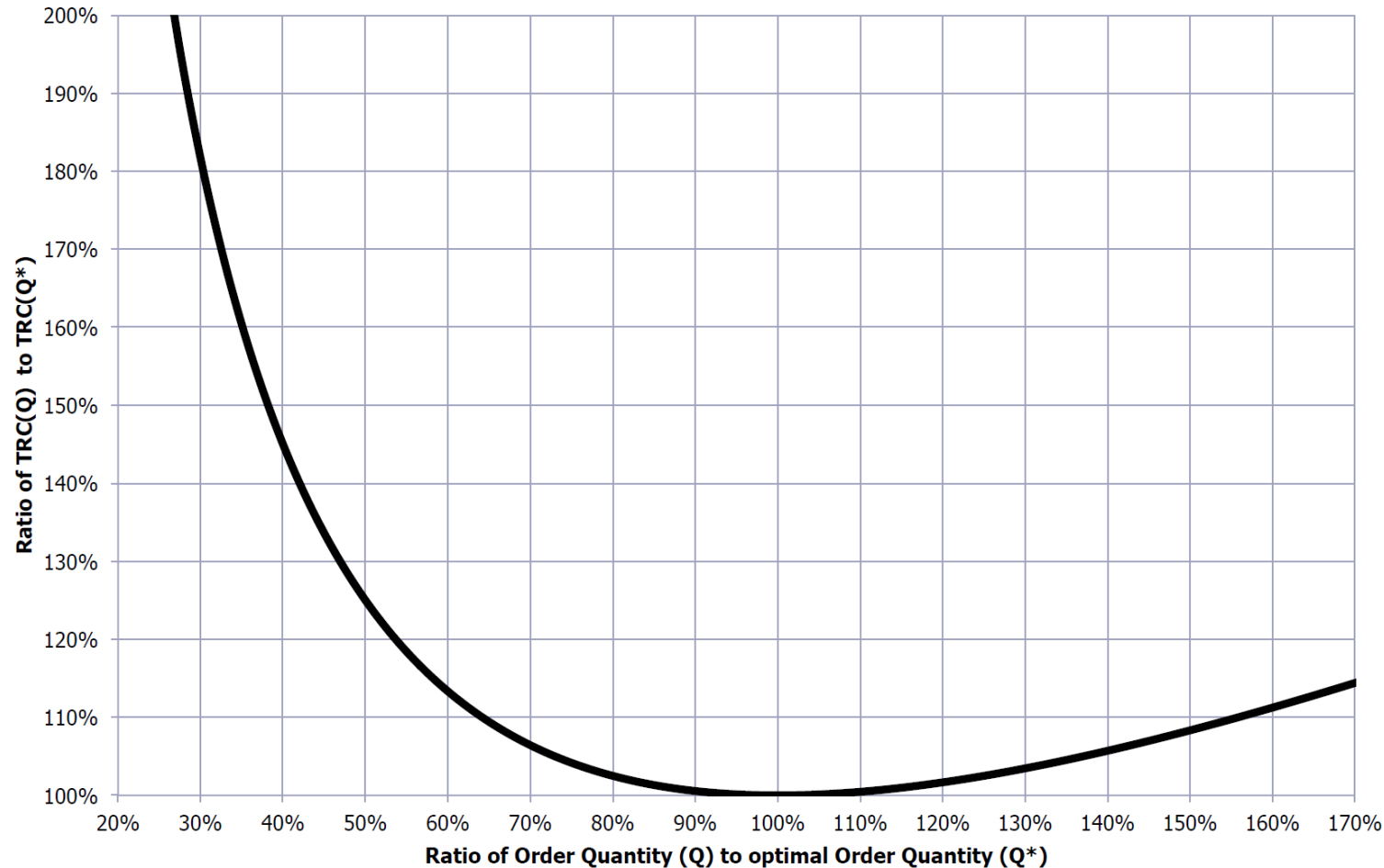
$$Z(Q)/Z(Q^*) = \frac{1}{2} (Q * \sqrt{h / (2 * A * D)} + 1/Q * \sqrt{2 * A * D / h})$$

$$Z(Q)/Z(Q^*) = \frac{1}{2} (Q/Q^* + Q^*/Q)$$

If Q/Q^* is 2 $Z(Q)/Z(Q^*)$ is 1,25.

A 100% error on Q leads to a 25% error on the variable cost.

EOQ model sensitivity: order quantity



Production Order Quantity model

During production the inventory builds up over the considered period, it is not received instantaneously as we assumed in EOQ. As we might be delivering to customer the inventory is being depleted at the same time.

Q: Number of units per production order

h: holding cost per unit per year

p: daily production rate

Dj: daily demand rate

D: yearly demand

t: length of the production run in days

We do not have an ordering cost but we have a set-up cost S.

Annual inventory holding cost = average inventory level * h

Average inventory level = (max inventory)/2

Maximum inventory level = (Total produced inventory during production run) – (total used during the production period)

$$\text{Maximum Inventory Level} = p * t - D_j * t$$

However: $Q = p * t$

$$\Rightarrow \text{Maximum Inventory Level} = Q * \left(1 - \frac{D_j}{p}\right)$$

Production Order Quantity model

$$\Rightarrow \text{Maximum Inventory Level} = Q * (1 - \frac{Dj}{p})$$

$$\Rightarrow \text{Annual inventory holding cost} = Q * h * (1 - \frac{Dj}{p})/2$$

$$\text{From the EOQ method we know that set-up cost is : } \text{Setup cost} = \frac{D}{Q} * S$$

The optimal Q is when setup cost equal holding cost

This gives us:

$$Q^* = \sqrt{\frac{2 S D}{h(1 - \frac{Dj}{p})}}$$

Agenda Course 3

- Inventory Planning for known demand
 - The role of inventory
 - Balance between different cost elements
 - Different inventory planning policies
 - EOQ sensitivity

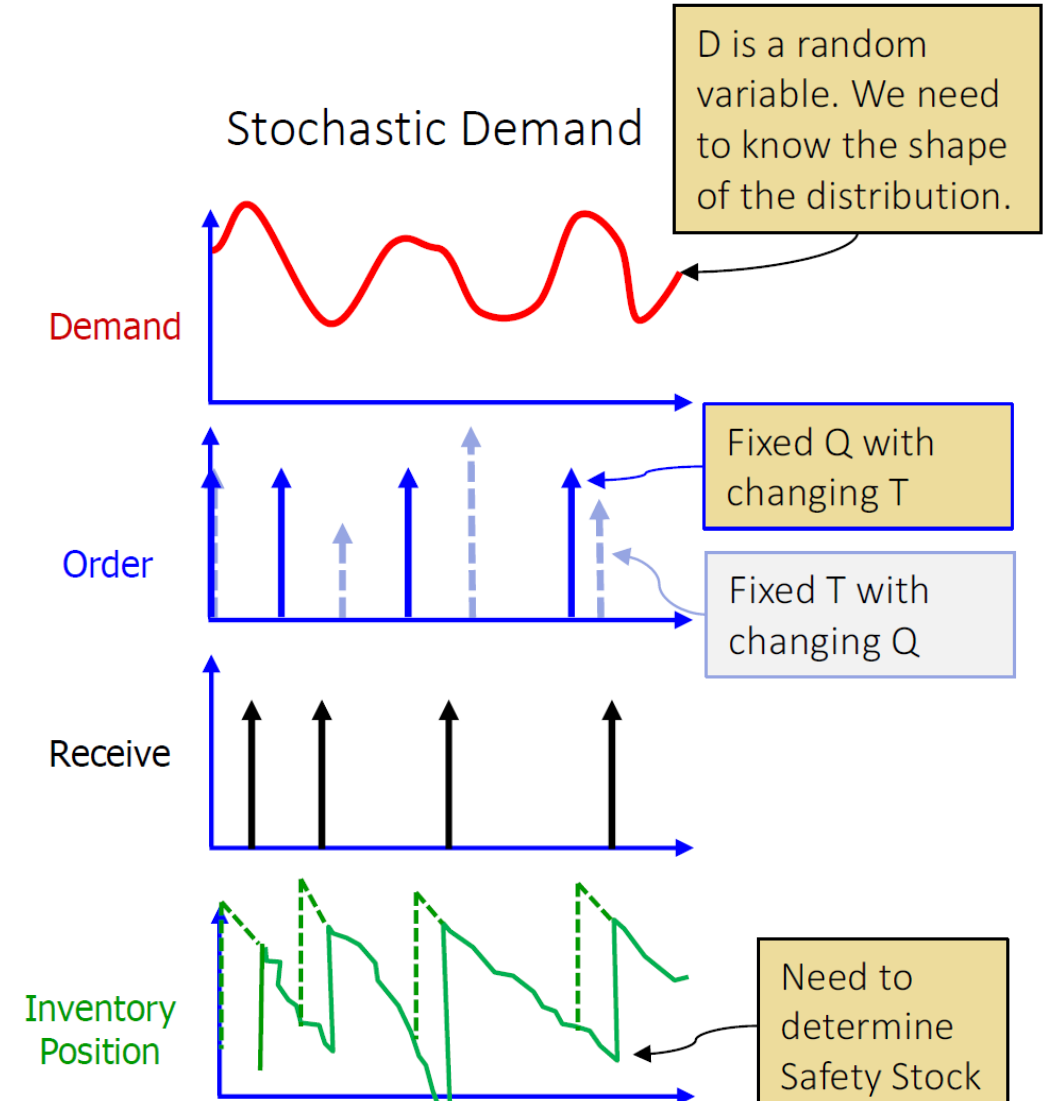
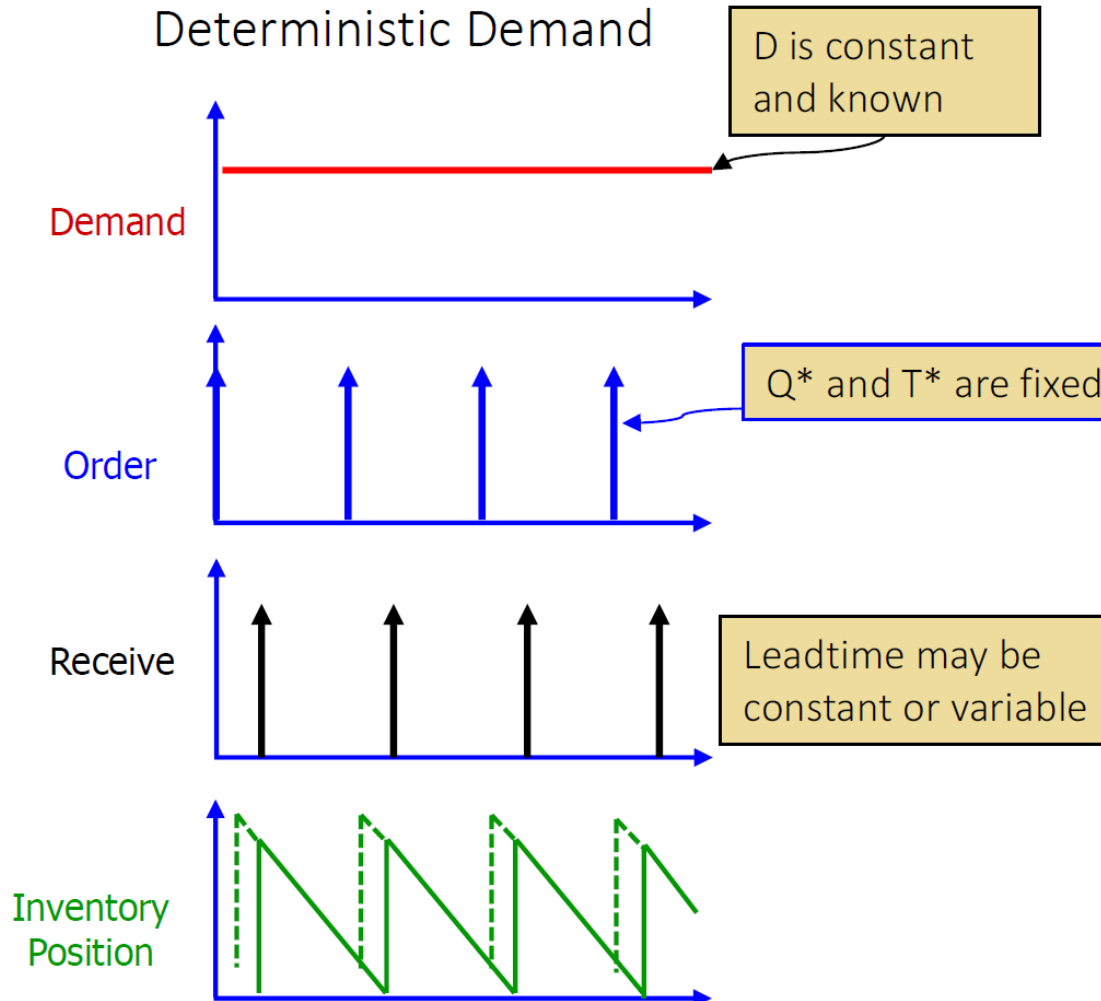
- Inventory Planning for unknown demand
 - Stochastic demand
 - News vendor model
 - Multi-echelon systems and bullwhip

Variable assumptions: stochastic demand

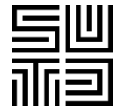


- Demand
 - Constant vs **Variable**
 - Known vs **Random**
 - **Continuous** vs Discrete
- Lead Time
 - Instantaneous
 - **Constant** vs Variable
 - Deterministic vs Stochastic
 - Internally Replenished
- Dependence of Items
 - **Independent**
 - Correlated
 - Indentured
- Review Time
 - **Continuous** vs Periodic
- Number of Locations
 - **One** vs Multi vs Multi-Echelon
- Capacity / Resources
 - **Unlimited**
 - Limited / Constrained
- Discounts
 - **None**
 - All Units vs Incremental vs One Time
- Excess Demand
 - None
 - All orders are backordered
 - **Lost orders**
 - Substitution
- Perishability
 - **None**
 - Uniform with time
 - Non-linear with time
- Planning Horizon
 - **Single Period**
 - Finite Period
 - **Infinite**
- Number of Items
 - **One** vs Many
- Form of Product
 - **Single Stage** vs Multi-Stage

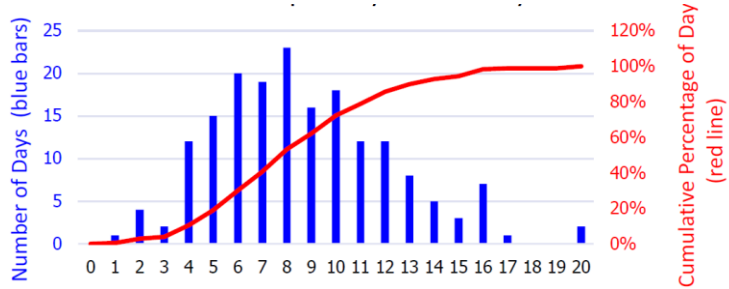
Deterministic vs stochastic demand: impact on the inventory



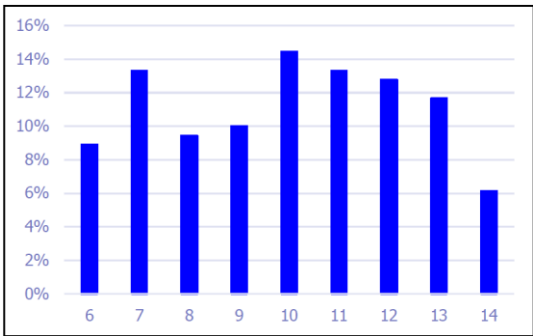
Demand Distribution



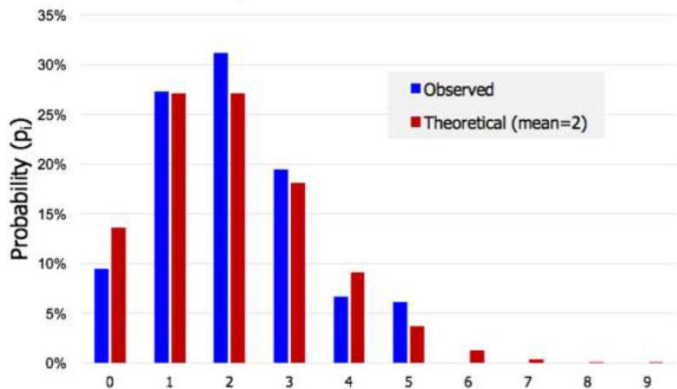
Discrete



Empirical

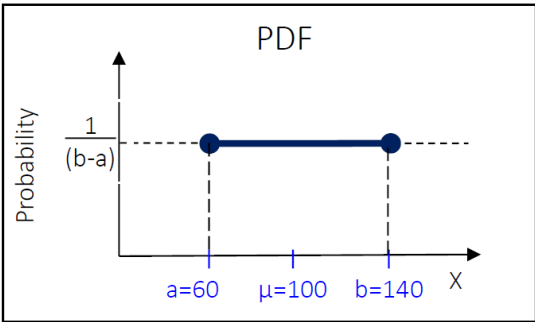


Discrete uniform

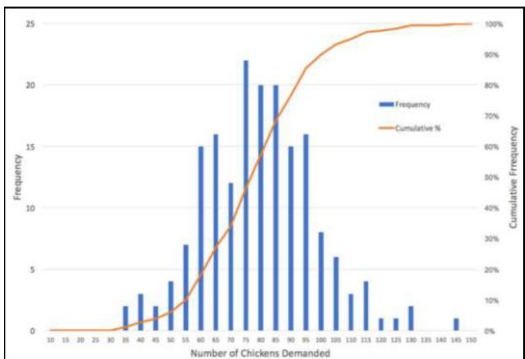


Poisson

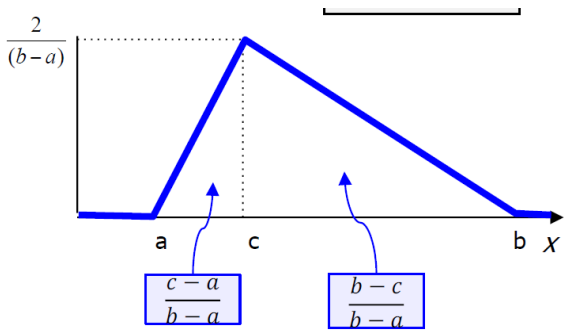
Continuous



Continuous uniform



Normal distribution



Triangle

Expected value for continuous distribution

We assume we have Q units in stock.

x is the demand and $f(x)$ the density function for that demand.

The expected demand is : **$E(x)$**

$$= \int_{x=0}^{\infty} x f_x(x) dx = \hat{x}$$

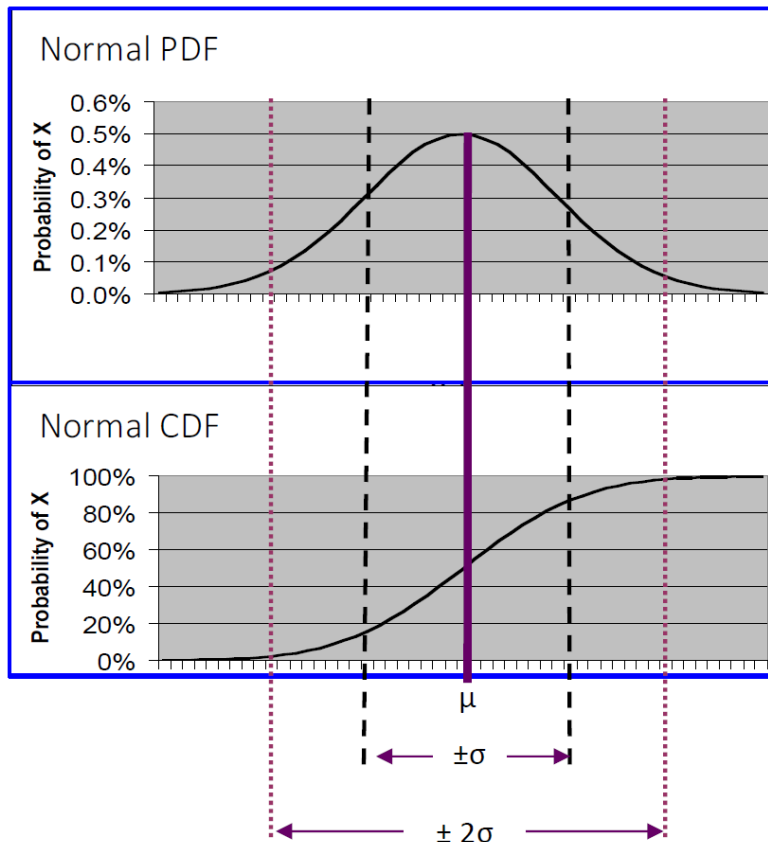
The expected units sold: **$E(\text{units sold})$** : once the demand is above Q you can only sell Q

$$= \int_{x=0}^Q x f_x(x) dx + Q \int_{x=Q}^{\infty} f_x(x) dx$$

The expected units short: **$E(\text{units shorts})$** :

$$= \int_{x=Q}^{\infty} (x - Q) f_x(x) dx$$

Refresher on normal distribution



Notation: $N(\mu, \sigma)$

μ = Mean

σ = Std Deviation

Common dispersion metrics $\sim N(\mu, \sigma)$

- $P(X \text{ w/in } 1\sigma \text{ around } \mu) = .6826$
- $P(X \text{ w/in } 2\sigma \text{ around } \mu) = .9544$
- $P(X \text{ w/in } 3\sigma \text{ around } \mu) = .9974$

- $\pm 1.65 \sigma \text{ around } \mu = 0.900$
- $\pm 1.96 \sigma \text{ around } \mu = 0.950$
- $\pm 2.81 \sigma \text{ around } \mu = 0.995$

68% of the probability falls within 1σ

To cover 90% we need to fall between 1,65 std dev

Note:

- The mean, μ , 'shifts' the distribution – but not the 'shape'
- The std. dev., σ , changes the 'shape' but doesn't 'shift' it

If we link to forecasting: the mean is influenced by the bias and the std dev is influenced by the accuracy.

Standard Normal distribution

All available tables are for the standard normal distribution. The mean is 0 and the std dev is 1.

We take all the value, subtract off the mean and divide by the standard deviation.

Standard Normal Distribution (k values)

- $\sim N(0,1)$ where the k value $= (X - \mu) / \sigma$
- The k value is simply the number of standard deviations away from the mean
- It can be positive or negative.
- k-values are widely used in inventory theory for setting safety stock
- Area under the Unit Normal curve is equal to 1

Function	Microsoft Excel & LibreOffice->Calc
CDF	<code>=NORM.DIST(X, μ, σ, 1)</code>
PDF	<code>=NORM.DIST(X, μ, σ, 0)</code>
Inverse CDF	<code>=NORM.INV(Probability, μ, σ)</code>
Standard Normal CDF	<code>=NORM.S.DIST(k,1)</code>
Inverse Standard Normal CDF	<code>=NORM.S.INV(Probability)</code>

Variable assumptions: single period



- Demand
 - Constant vs **Variable**
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News vendor model

We assume a normal distribution for the demand (density function $f(x)$ and cumulative density function $F(x)$) and a single planning horizon.

We will look at the inventory to order in order to minimize cost.

The name comes from the boys selling newspaper on the street. They have to decide how many newspaper to buy. If they overstock they have to scrap their inventory as the newspaper is obsolete the following day.

The specificity of a single period model is that the selling period is limited and at the end we cannot sell the product at full price. During that period we CANNOT replenish the inventory.

Newsvendor model

We have 2 costs:

- Shortage cost if the demand exceeds the inventory Q : C_u
- Excess cost if the demand is lower than Q : C_o

If our buying price is c and our selling price is p it turns that:

$$C_o = c$$

$$C_u = p - c$$

We want to:

Develop expression for cost as function of D and Q : $G(D, Q)$

Determine expected value of cost function

Determine Q^* that minimizes expected cost

It follows:

$$G(Q, D) = \begin{cases} c_o(Q - D) & \text{if } Q > D \\ c_u(D - Q) & \text{for } Q < D \end{cases}$$

Newsvendor model

It can be formulated as:

$$G(Q,D) = C_o * \max(0, Q-D) + C_u * \max(0, D-Q)$$

Considering we have a normal distribution for the demand the expected cost will be:

$$G(Q) = E(G(Q,D))$$

$$\begin{aligned} G(Q) &= c_o \int_0^{\infty} \max(0, Q-x) f(x) dx \\ &\quad + c_u \int_0^{\infty} \max(0, x-Q) f(x) dx \\ &= c_o \int_0^Q (Q-x) f(x) dx \\ &\quad + c_u \int_Q^{\infty} (x-Q) f(x) dx \end{aligned}$$

Newsvendor model

We want to minimize the expected cost. Using Leibniz rule it follows that:

$$\begin{aligned}\frac{dG(Q)}{dQ} &= c_o \int_0^Q 1f(x)dx + c_u \int_Q^\infty (-1)f(x)dx \\ &= c_o F(Q) - c_u (1 - F(Q))\end{aligned}$$

It follows that

$$\frac{d^2 G(Q)}{dQ^2} = (c_o + c_u)f(Q) \geq 0 \text{ for all } Q \geq 0$$

The function is *convex*

$$\begin{aligned}G'(Q^*) &= (c_o + c_u)F(Q^*) - c_u = 0 \\ F(Q^*) &= \frac{c_u}{(c_o + c_u)}\end{aligned}$$

The **Critical ratio CR** is : $CR = C_u / (C_o + C_u)$.

We can find Q^* using a normal distribution table ($F(k)$) or xls formula, considering $k = (Q - \mu) / \sigma$

Example

In our newspaper case considering:

- Purchase cost c of SGD0,35
- Selling price of SGD0,75
- Demand distribution $N(15,3,0)$

$$\Rightarrow C_o = c = 0,35$$

$$\Rightarrow C_u = 0,75 - 0,35 = 0,4$$

$$\Rightarrow CR = 0,53$$

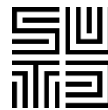
$$\Rightarrow k = 0,075 \text{ (looking at the table)}$$

$$\Rightarrow Q^* = 3 \cdot 0,075 + 15 = 15,22$$

Or

$$\Rightarrow Q^* = \text{NORMINV}(CR, \text{Mean}, \text{StdDev})$$

k	P[x≤k]	G(k)	k	P[x≤k]	G(k)	k	P[x≤k]
0.00	0.5000	0.3989	0.50	0.6915	0.1978	1.00	0.84
0.01	0.5040	0.3940	0.51	0.6950	0.1947	1.01	0.84
0.02	0.5080	0.3890	0.52	0.6985	0.1917	1.02	0.84
0.03	0.5120	0.3841	0.53	0.7019	0.1887	1.03	0.84
0.04	0.5160	0.3793	0.54	0.7054	0.1857	1.04	0.85
0.05	0.5199	0.3744	0.55	0.7088	0.1828	1.05	0.85
0.06	0.5239	0.3697	0.56	0.7123	0.1799	1.06	0.85
0.07	0.5279	0.3649	0.57	0.7157	0.1771	1.07	0.85
0.08	0.5319	0.3602	0.58	0.7190	0.1742	1.08	0.85
0.09	0.5359	0.3556	0.59	0.7224	0.1714	1.09	0.86
0.10	0.5398	0.3509	0.60	0.7257	0.1687	1.10	0.86
0.11	0.5438	0.3464	0.61	0.7291	0.1659	1.11	0.86
0.12	0.5478	0.3418	0.62	0.7324	0.1633	1.12	0.86
0.13	0.5517	0.3373	0.63	0.7357	0.1606	1.13	0.87
0.14	0.5557	0.3328	0.64	0.7389	0.1580	1.14	0.87
0.15	0.5596	0.3284	0.65	0.7422	0.1554	1.15	0.87
0.16	0.5636	0.3240	0.66	0.7454	0.1528	1.16	0.87
0.17	0.5675	0.3197	0.67	0.7486	0.1503	1.17	0.87
0.18	0.5714	0.3154	0.68	0.7517	0.1478	1.18	0.88
0.19	0.5753	0.3111	0.69	0.7549	0.1453	1.19	0.88
0.20	0.5793	0.3069	0.70	0.7580	0.1429	1.20	0.88
0.21	0.5832	0.3027	0.71	0.7611	0.1405	1.21	0.88



Example: what happens if we can sell the newspaper at a discount

The shortage cost is the same.

We now have a **salvage value** g .

The excess cost becomes: $C_o = c - g$.

The CR becomes:

$$CR(\text{salvage}) = C_u / (c - g + C_u)$$

$$CR(\text{salvage}) > CR(\text{standard})$$

⇒ I can buy an higher quantity of material.

⇒ My revenue will be higher

Profit maximization

The profit can be expressed as:

$$P(Q, x) : \begin{array}{ll} (p^*x - c^*Q) & \text{if } x \leq Q \\ (p^*Q - c^*Q) & \text{if } x > Q \end{array}$$

$$E(P(Q)) = \int_0^{\infty} P(Q, x) f(x) dx = \int_0^Q (px - cQ) f(x) dx + \int_Q^{\infty} (pQ - cQ) f(x) dx$$

$$E(P(Q)) = p \int_0^{\infty} xf(x) dx - cQ - p \int_Q^{\infty} (x - Q) f(x) dx$$

$$E(P(Q)) = pE(x) - cQ - pE(\text{unitsshort})$$

Expected Units shorts

$$E(\text{unitsshort}) = \int_Q^{\infty} (x - Q)f(x)dx$$

We introduce the loss function designed for the unit standard distribution $G(k)$. It gives the units shorts for the unit normal curve as a fraction of the standard deviation.

We scale to our distribution which has a standard deviation of σ .

So:

$$E(\text{unitsshort}) = \sigma G\left(\frac{Q - \mu}{\sigma}\right)$$

Recall that k is the Standard Normal Distribution where ($\mu=0, \sigma=1$)
 Transform: $k=(Q - \mu)/\sigma$ In Spreadsheets: $k=\text{NORMSINV}(\text{CR})$

Unit Normal Loss function is:

$$G(k) = f_x(x_0) - k \cdot \text{Prob}[x_0 \geq k]$$

In Spreadsheets:

$$=\text{NORMDIST}(\mathbf{k}, 0, 1, 0) - \mathbf{k} * (1 - \text{NORMSDIST}(\mathbf{k}))$$

Expected Units shorts using Table

Suppose:

$\sim N(160, 45)$

and $Q = 190$

What is my $E[US]$?

$$k = (190 - 160) / 45 \\ = 0.67$$

$$G(k) = 0.1503$$

$$E[US] = (45)(0.1503) \\ = 6.76 \text{ units}$$

k	P[x≤k]	G(k)	k	P[x≤k]	G(k)	k	P[x≤k]	G(k)
0.00	0.5000	0.3989	0.50	0.6915	0.1978	1.00	0.8413	0.0833
0.01	0.5040	0.3940	0.51	0.6950	0.1947	1.01	0.8438	0.0817
0.02	0.5080	0.3890	0.52	0.6985	0.1917	1.02	0.8461	0.0802
0.03	0.5120	0.3841	0.53	0.7019	0.1887	1.03	0.8485	0.0787
0.04	0.5160	0.3793	0.54	0.7054	0.1857	1.04	0.8508	0.0772
0.05	0.5199	0.3744	0.55	0.7088	0.1828	1.05	0.8531	0.0757
0.06	0.5239	0.3697	0.56	0.7123	0.1799	1.06	0.8554	0.0742
0.07	0.5279	0.3649	0.57	0.7157	0.1771	1.07	0.8577	0.0728
0.08	0.5319	0.3602	0.58	0.7190	0.1742	1.08	0.8599	0.0714
0.09	0.5359	0.3556	0.59	0.7224	0.1714	1.09	0.8621	0.0700
0.10	0.5398	0.3509	0.60	0.7257	0.1687	1.10	0.8643	0.0686
0.11	0.5438	0.3464	0.61	0.7291	0.1659	1.11	0.8665	0.0673
0.12	0.5478	0.3418	0.62	0.7324	0.1633	1.12	0.8686	0.0659
0.13	0.5517	0.3373	0.63	0.7357	0.1606	1.13	0.8708	0.0646
0.14	0.5557	0.3328	0.64	0.7389	0.1580	1.14	0.8729	0.0634
0.15	0.5596	0.3284	0.65	0.7422	0.1554	1.15	0.8749	0.0621
0.16	0.5636	0.3240	0.66	0.7454	0.1528	1.16	0.8770	0.0609
0.17	0.5675	0.3197	0.67	0.7486	0.1503	1.17	0.8790	0.0596
0.18	0.5714	0.3154	0.68	0.7517	0.1478	1.18	0.8810	0.0584
0.19	0.5753	0.3111	0.69	0.7549	0.1453	1.19	0.8830	0.0573
0.20	0.5793	0.3069	0.70	0.7580	0.1429	1.20	0.8849	0.0561
0.21	0.5832	0.3027	0.71	0.7611	0.1405	1.21	0.8869	0.0550

Expected Units shorts for our newspaper case

Suppose:

$\sim N(15, 3)$

$k=0,075$

What is my $E[US]$?

$G(k) = 0,362$

$E[US] = (3)(0.362)$
 $= 1.08$ units

k	P[x≤k]	G(k)	k	P[x≤k]	G(k)	k	P[x≤k]	G(k)
0.00	0.5000	0.3989	0.50	0.6915	0.1978	1.00	0.8413	0.0833
0.01	0.5040	0.3940	0.51	0.6950	0.1947	1.01	0.8438	0.0817
0.02	0.5080	0.3890	0.52	0.6985	0.1917	1.02	0.8461	0.0802
0.03	0.5120	0.3841	0.53	0.7019	0.1887	1.03	0.8485	0.0787
0.04	0.5160	0.3793	0.54	0.7054	0.1857	1.04	0.8508	0.0772
0.05	0.5199	0.3744	0.55	0.7088	0.1828	1.05	0.8531	0.0757
0.06	0.5239	0.3697	0.56	0.7123	0.1799	1.06	0.8554	0.0742
0.07	0.5279	0.3649	0.57	0.7157	0.1771	1.07	0.8577	0.0728
0.08	0.5319	0.3602	0.58	0.7190	0.1742	1.08	0.8599	0.0714
0.09	0.5359	0.3556	0.59	0.7224	0.1714	1.09	0.8621	0.0700
0.10	0.5398	0.3509	0.60	0.7257	0.1687	1.10	0.8643	0.0686
0.11	0.5438	0.3464	0.61	0.7291	0.1659	1.11	0.8665	0.0673
0.12	0.5478	0.3418	0.62	0.7324	0.1633	1.12	0.8686	0.0659
0.13	0.5517	0.3373	0.63	0.7357	0.1606	1.13	0.8708	0.0646
0.14	0.5557	0.3328	0.64	0.7389	0.1580	1.14	0.8729	0.0634
0.15	0.5596	0.3284	0.65	0.7422	0.1554	1.15	0.8749	0.0621
0.16	0.5636	0.3240	0.66	0.7454	0.1528	1.16	0.8770	0.0609
0.17	0.5675	0.3197	0.67	0.7486	0.1503	1.17	0.8790	0.0596
0.18	0.5714	0.3154	0.68	0.7517	0.1478	1.18	0.8810	0.0584
0.19	0.5753	0.3111	0.69	0.7549	0.1453	1.19	0.8830	0.0573
0.20	0.5793	0.3069	0.70	0.7580	0.1429	1.20	0.8849	0.0561
0.21	0.5832	0.3027	0.71	0.7611	0.1405	1.21	0.8869	0.0550

Example

We buy tumblers for a sport event. We order prior the event, cannot place orders during the event. Whatever is not sold will be scrapped.

The buying price is 10,9SGD/tumbler.

The selling price is 24SGD/tumbler.

Based on experience we expect the demand to be normally distributed with $N(32000, 11000)$.

Which quantity to order to optimize profit?

Example

Which quantity to order to optimize profit?

$$CR = (p - c) / p = 0,546$$

$$\Rightarrow k = 0,115$$

$$k = (Q^* - 32000) / 11000$$

$$Q^* = 33267 \text{ units}$$

What is the profit?

$$E(P(Q)) = p \cdot E(x) - cQ - p \cdot E(\text{unitsshort})$$

$$p \cdot E(x) = 24 \cdot 32000 = 768000 \text{ SGD}$$

$$cQ = 10,9 \cdot 33267 = 362610 \text{ SGD}$$

$$G(k) = 0,3441$$

$$E(\text{unitsshort}) = 11000 \cdot G(k) = 3785$$

$$p \cdot E(\text{unitsshort}) = 90840$$

$$E(P(Q)) = 768000 - 362610 - 90840 = 314550 \text{ SGD}$$

k	P[x≤k]	G(k)	k	P[x≤k]	G(k)	k	P[x≤k]	G(k)
0.00	0.5000	0.3989	0.50	0.6915	0.1978	1.00	0.8413	0.0833
0.01	0.5040	0.3940	0.51	0.6950	0.1947	1.01	0.8438	0.0817
0.02	0.5080	0.3890	0.52	0.6985	0.1917	1.02	0.8461	0.0802
0.03	0.5120	0.3841	0.53	0.7019	0.1887	1.03	0.8485	0.0787
0.04	0.5160	0.3793	0.54	0.7054	0.1857	1.04	0.8508	0.0772
0.05	0.5199	0.3744	0.55	0.7088	0.1828	1.05	0.8531	0.0757
0.06	0.5239	0.3697	0.56	0.7123	0.1799	1.06	0.8554	0.0742
0.07	0.5279	0.3649	0.57	0.7157	0.1771	1.07	0.8577	0.0728
0.08	0.5319	0.3602	0.58	0.7190	0.1742	1.08	0.8599	0.0714
0.09	0.5359	0.3556	0.59	0.7224	0.1714	1.09	0.8621	0.0700
0.10	0.5398	0.3509	0.60	0.7257	0.1687	1.10	0.8643	0.0686
0.11	0.5438	0.3464	0.61	0.7291	0.1659	1.11	0.8665	0.0673
0.12	0.5478	0.3418	0.62	0.7324	0.1633	1.12	0.8686	0.0659
0.13	0.5517	0.3373	0.63	0.7357	0.1606	1.13	0.8708	0.0646
0.14	0.5557	0.3328	0.64	0.7389	0.1580	1.14	0.8729	0.0634
0.15	0.5596	0.3284	0.65	0.7422	0.1554	1.15	0.8749	0.0621
0.16	0.5636	0.3240	0.66	0.7454	0.1528	1.16	0.8770	0.0609
0.17	0.5675	0.3197	0.67	0.7486	0.1503	1.17	0.8790	0.0596
0.18	0.5714	0.3154	0.68	0.7517	0.1478	1.18	0.8810	0.0584
0.19	0.5753	0.3111	0.69	0.7549	0.1453	1.19	0.8830	0.0573
0.20	0.5793	0.3069	0.70	0.7580	0.1429	1.20	0.8849	0.0561
0.21	0.5832	0.3027	0.71	0.7611	0.1405	1.21	0.8869	0.0550

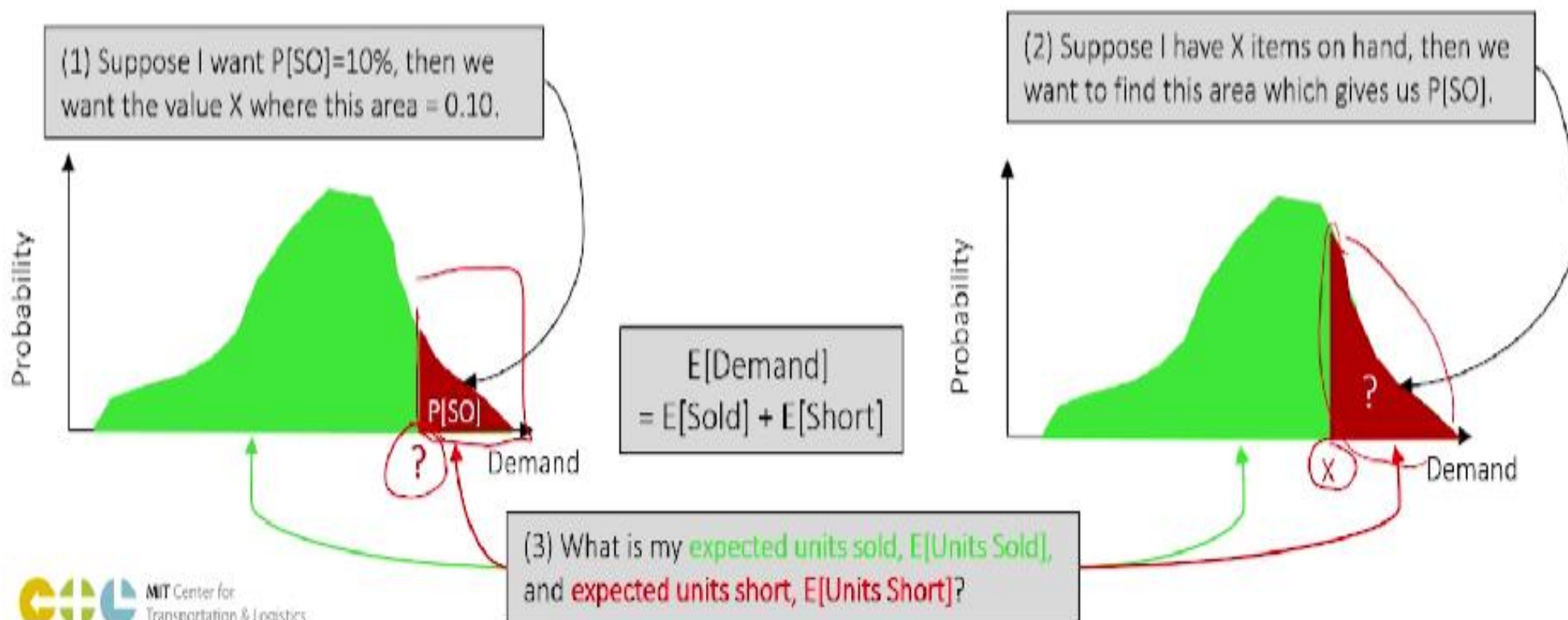
Variable assumptions: stochastic demand/infinite

- Demand
 - Constant vs **Variable**
 - Known vs **Random**
 - **Continuous** vs Discrete
- Lead Time
 - Instantaneous
 - **Constant** vs Variable
 - Deterministic vs Stochastic
 - Internally Replenished
- Dependence of Items
 - **Independent**
 - Correlated
 - Indentured
- Review Time
 - **Continuous** vs Periodic
- Number of Locations
 - **One** vs Multi vs Multi-Echelon
- Capacity / Resources
 - **Unlimited**
 - Limited / Constrained
- Discounts
 - **None**
 - All Units vs Incremental vs One Time
- Excess Demand
 - None
 - All orders are backordered
 - **Lost orders**
 - Substitution
- Perishability
 - **None**
 - Uniform with time
 - Non-linear with time
- Planning Horizon
 - **Single Period**
 - Finite Period
 - **Infinite**
- Number of Items
 - **One** vs Many
- Form of Product
 - **Single Stage** vs Multi-Stage

Solving for service level

Assuming a demand distribution:

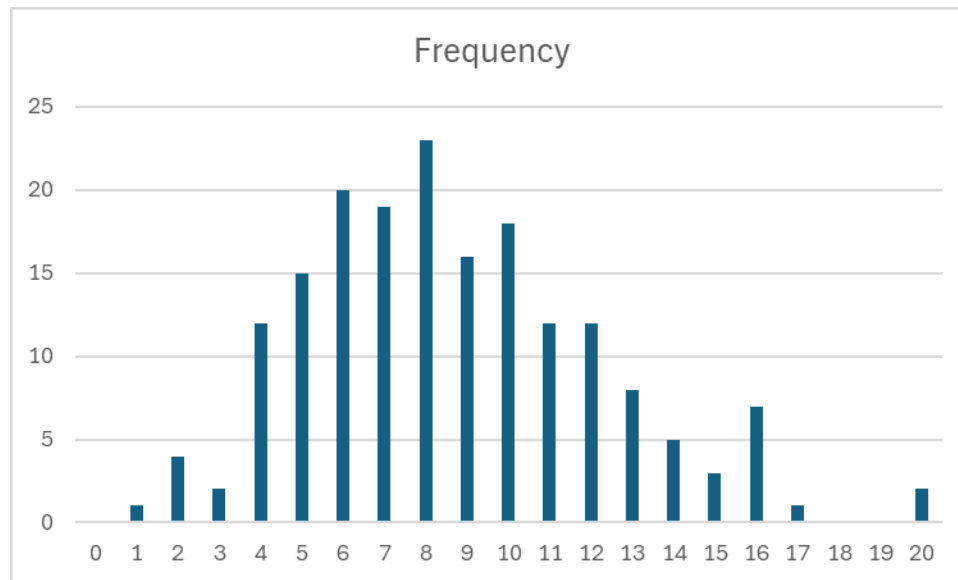
- How much inventory do I need to carry to guarantee a service level?
- If I have inventory what service level will it guarantee?
- Given a targeted service level or inventory on hand how many units am I expected to sell?



Empirical distribution

Remembering our cookies business we have recorded below demand over 180 days:

Cookies batch	Frequency	Probability	Cumulative Probability
0	0	0,0%	0,0%
1	1	0,6%	0,6%
2	4	2,2%	2,8%
3	2	1,1%	3,9%
4	12	6,7%	10,6%
5	15	8,3%	18,9%
6	20	11,1%	30,0%
7	19	10,6%	40,6%
8	23	12,8%	53,3%
9	16	8,9%	62,2%
10	18	10,0%	72,2%
11	12	6,7%	78,9%
12	12	6,7%	85,6%
13	8	4,4%	90,0%
14	5	2,8%	92,8%
15	3	1,7%	94,4%
16	7	3,9%	98,3%
17	1	0,6%	98,9%
18	0	0,0%	98,9%
19	0	0,0%	98,9%
20	2	1,1%	100,0%



$$\mu = \sum_{i=1}^n p_i x_i$$

μ = Mean = Expected Value
= $E[X]$ = Average = 8.68

$$\sigma = \sqrt{\sum_{i=1}^n p_i (x_i - \mu)^2} = \sqrt{E[X^2] - E[X]^2}$$

Standard Deviation = 3.59 Batches/day

Empirical distribution



Remembering our cookies business we have recorded below demand over 180 days:

Cookies batch	Frequency	Probability	Cumulative Probability
0	0	0,0%	0,0%
1	1	0,6%	0,6%
2	4	2,2%	2,8%
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4	12	6,7%	10,6%
5	15	8,3%	18,9%
6	20	11,1%	30,0%
7	19	10,6%	40,6%
8	23	12,8%	53,3%
9	16	8,9%	62,2%
10	18	10,0%	72,2%
11	12	6,7%	78,9%
12	12	6,7%	85,6%
13	8	4,4%	90,0%
14	5	2,8%	92,8%
15	3	1,7%	94,4%
16	7	3,9%	98,3%
17	1	0,6%	98,9%
18	0	0,0%	98,9%
19	0	0,0%	98,9%
20	2	1,1%	100,0%

How many batches do we need to prepare to offer a 90% service level?
We use the cumulative probability to cover 90% of the demand.
Thus **13 batches**.

We have produced 8 batches as we had to review our exam paper. What is the probability of stocking out?
46,7%

What are expected batches sold if we have 8 batches:

- When demand is less than 8 we will sell what is demanded
- When demand is more than 8 we will sell 8

$$E(\text{Batches Sold}) = p_0 \cdot 0 + p_1 \cdot 1 + p_2 \cdot 2 + \dots + p_8 \cdot 8 + p_9 \cdot 8 + \dots + p_{20} \cdot 8$$

$$E(\text{Batches Sold}) = 6,93 \text{ batches/day}$$

$$E(\text{batches short}) = p_0 \cdot 0 + p_1 \cdot 0 + \dots + p_9 \cdot (9-8) + p_{10} \cdot (10-8) + \dots + p_{20} \cdot (20-8)$$

$$E(\text{batches short}) = 1,75 \text{ batches/day}$$

Note that : $E(\text{demand}) = E(\text{batches sold}) + E(\text{batches short})$

Normal Distribution demand

We have scaled up our business and assume the demand is normally distributed with a mean of 78 and a std dev of 19.

How many batches do we need to prepare to offer a 90% service level?

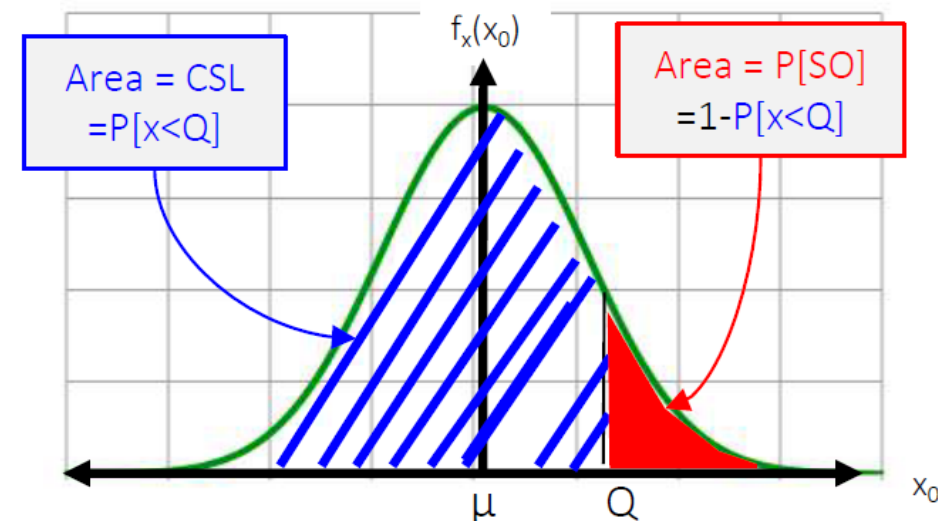
Use the inverse Normal function in your spreadsheet.

$\text{NORM.INV}(0.90, 78, 19) = 102.3 \Rightarrow 102$ batches of cookies

We have produced 99 batches. What is the probability of stocking out?

Using the cumulative distribution, $=\text{NORM.DIST}(99, 78, 19, 1) = 0.87$

So, since $P[\text{SO}] = 1 - \text{CSL} = 1 - 0.87 = 0.13$ or $\sim 13\%$



What are expected batches sold and shorts if we have produced 99 batches:

The $E[\text{units short}]$ follows the same formula as for the newsvendor model: $\sigma G(k)$

Using our standard table we have:

$k = (99 - 78) / 19 = 1.105$

$G(k) = 0.06795$

$E(\text{short}) = 19 * 0.06795 = 1.3$ batches

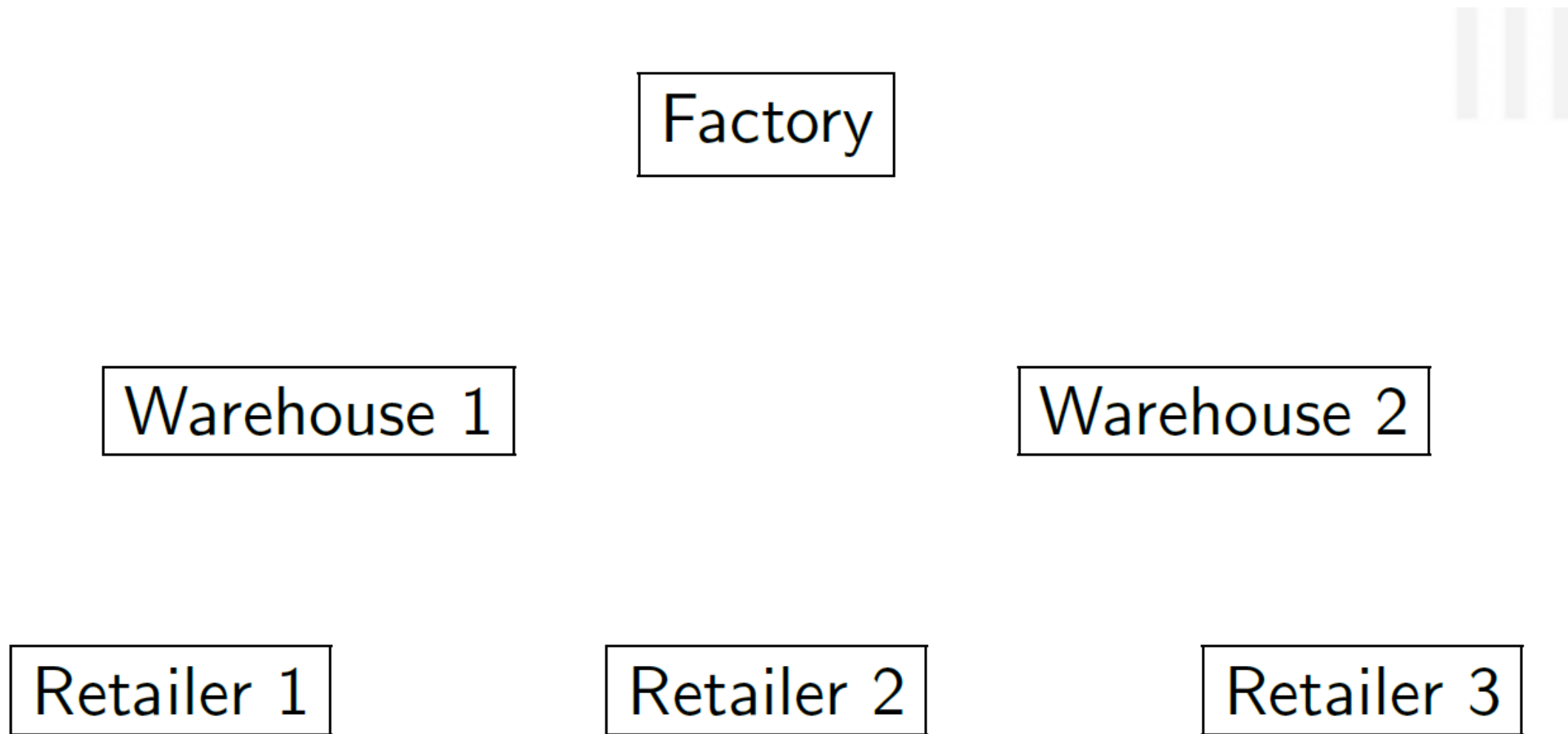
The expected demand is 78.

The expected batches sold are: $78 - 1.3 = 76.7$.

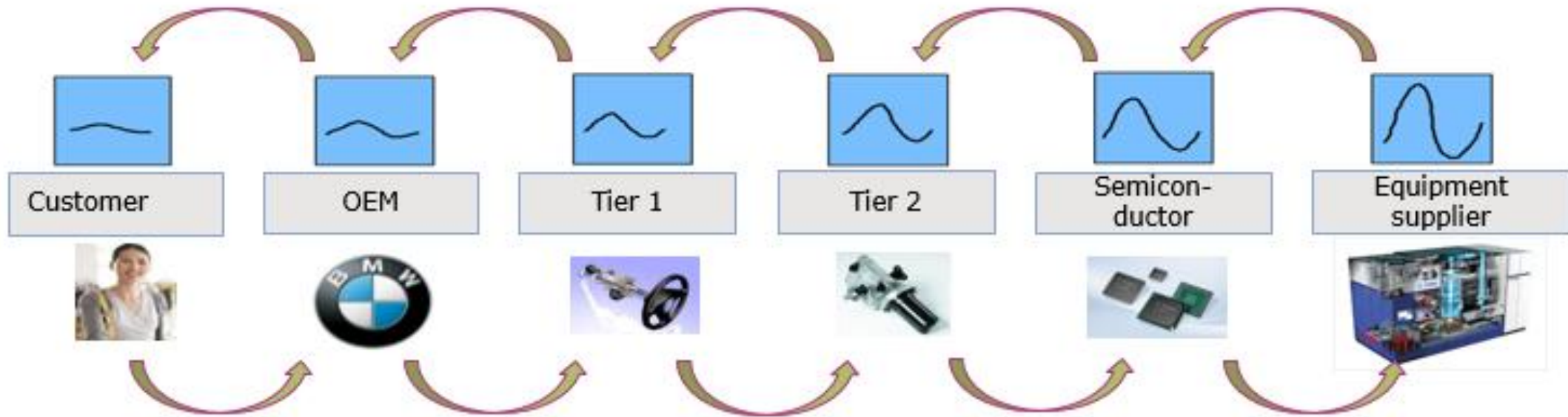
If we produce 99 batches we are expecting to sell 76,7 and be short of 1,3.

k	P[x≤k]	G(k)	k	P[x≤k]	G(k)	k	P[x≤k]	G(k)
0.00	0.5000	0.3989	0.50	0.6915	0.1978	1.00	0.8413	0.0833
0.01	0.5040	0.3940	0.51	0.6950	0.1947	1.01	0.8438	0.0817
0.02	0.5080	0.3890	0.52	0.6985	0.1917	1.02	0.8461	0.0802
0.03	0.5120	0.3841	0.53	0.7019	0.1887	1.03	0.8485	0.0787
0.04	0.5160	0.3793	0.54	0.7054	0.1857	1.04	0.8508	0.0772
0.05	0.5199	0.3744	0.55	0.7088	0.1828	1.05	0.8531	0.0757
0.06	0.5239	0.3697	0.56	0.7123	0.1799	1.06	0.8554	0.0742
0.07	0.5279	0.3649	0.57	0.7157	0.1771	1.07	0.8577	0.0728
0.08	0.5319	0.3602	0.58	0.7190	0.1742	1.08	0.8599	0.0714
0.09	0.5359	0.3556	0.59	0.7224	0.1714	1.09	0.8621	0.0700
0.10	0.5398	0.3509	0.60	0.7257	0.1687	1.10	0.8643	0.0686
0.11	0.5438	0.3464	0.61	0.7291	0.1659	1.11	0.8665	0.0673
0.12	0.5478	0.3418	0.62	0.7324	0.1633	1.12	0.8686	0.0659

Multi-echelon systems



Bullwhip



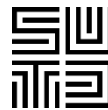
Effect of inventory on bullwhip

Without collaboration and 10% safety stock



With collaboration and 10% safety stock





Inventory game

Summary

Inventory is a cost item but needed for flexibility and risk management.

Various types of inventory.

Inventory can be smartly positioned along the supply chain.

ABC analysis to decide upon which inventory policy to use for which item.

Economic Order Quantity concept for independent demand

ReOrder Point method or Periodic Replenishment are inventory policies considering that the replenishment lead time is not instantaneous.

The News Vendor model caters for single period inventory decision for random demand.

THANK YOU

TRAILBLAZING A BETTER WORLD BY DESIGN.

